

AWS Certified Solutions Architect - Associate SAA-C03

question 0301-0400

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【題號】question 301

【題目】

A university research laboratory needs to migrate 30 TB of data from an on-premises Windows file server to Amazon FSx for Windows File Server. The laboratory has a 1 Gbps network link that many other departments in the university share.

The laboratory wants to implement a data migration service that will maximize the performance of the data transfer. However, the laboratory needs to be able to control the amount of bandwidth that the service uses to minimize the impact on other departments. The data migration must take place within the next 5 days.

Which AWS solution will meet these requirements?

某大學研究實驗室需要將 30 TB 資料從本機 Windows 檔案伺服器移轉到 Amazon FSx for Windows File Server。該實驗室擁有 1 Gbps 的網路鏈路，與大學的許多其他部門共用。

該實驗室希望實施一項資料遷移服務，以最大限度地提高資料傳輸的效能。然而，實驗室需要能夠控制服務使用的頻寬量，以盡量減少對其他部門的影響。資料遷移必須在接下來的 5 天內進行。

哪種 AWS 解決方案可以滿足這些要求？

【選項】

- A.AWS Snowcone
- B.Amazon FSx File Gateway
- C.AWS DataSync
- D.AWS Transfer Family

【參考答案】C

【題號】question 302

【題目】

A company wants to create a mobile app that allows users to stream slow-motion video clips on their mobile devices. Currently, the app captures video clips and uploads the video clips in raw format into an Amazon S3 bucket. The app retrieves these video clips directly from the S3 bucket. However, the videos are large in their raw format.

Users are experiencing issues with buffering and playback on mobile devices. The company wants to implement solutions to maximize the performance and scalability of the app while minimizing operational overhead.

Which combination of solutions will meet these requirements? (Choose two.)

一家公司想要創建一個行動應用程序，允許用戶在其行動裝置上串流慢動作影片剪輯。目前，該應用程式會擷取影片剪輯並將原始格式的影片剪輯上傳到 Amazon S3 儲存桶中。該應用程式直接從 S3 儲存桶檢索這些影片剪輯。然而，這些影片的原始格式很大。

用戶在行動裝置上遇到緩衝和播放問題。該公司希望實施解決方案，以最大限度地提高應用程式的效能和可擴展性，同時最大限度地減少營運開銷。

哪種解決方案組合可以滿足這些要求？（選兩個。）

【選項】

A. Deploy Amazon CloudFront for content delivery and caching.

B. Use AWS DataSync to replicate the video files across AWS Regions in other S3 buckets.

C. Use Amazon Elastic Transcoder to convert the video files to more appropriate formats.

D. Deploy an Auto Scaling group of Amazon EC2 instances in Local Zones for content delivery and caching.

E. Deploy an Auto Scaling group of Amazon EC2 instances to convert the video files to more appropriate formats.

【參考答案】A

【題號】question 303

【題目】

A company is launching a new application deployed on an Amazon Elastic Container Service (Amazon ECS) cluster and is using the Fargate launch type for ECS tasks. The company is monitoring CPU and memory usage because it is expecting high traffic to the application upon its launch. However, the company wants to reduce costs when utilization decreases.

What should a solutions architect recommend?

一家公司正在啟動部署在 Amazon Elastic Container Service (Amazon ECS) 叢集上的新應用程式，並使用 Fargate 啟動類型來執行 ECS 任務。該公司正在監控 CPU 和記憶體使用情況，因為預計應用程式啟動後會有大量流量。然而，該公司希望在利用率下降時降低成本。

解決方案架構師該推薦什麼？

【選項】

A. Use Amazon EC2 Auto Scaling to scale at certain periods based on previous traffic patterns.

B. Use an AWS Lambda function to scale Amazon ECS based on metric breaches that trigger an Amazon CloudWatch alarm.

C. Use Amazon EC2 Auto Scaling with simple scaling policies to scale when ECS metric breaches trigger an Amazon CloudWatch alarm.

D. Use AWS Application Auto Scaling with target tracking policies to scale when ECS metric breaches trigger an Amazon CloudWatch alarm.

【參考答案】D

【題號】question 304

【題目】

A company recently created a disaster recovery site in a different AWS Region. The company needs to transfer large amounts of data back and forth between NFS file systems in the two Regions on a periodic basis.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司最近在不同的 AWS 區域建立了一個災難復原站點。該公司需要定期在兩個區域的 NFS 檔案系統之間來回傳輸大量資料。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

- A. Use AWS DataSync.
- B. Use AWS Snowball devices.
- C. Set up an SFTP server on Amazon EC2.
- D. Use AWS Database Migration Service (AWS DMS).

【參考答案】A

【題號】question 305

【題目】

A company is designing a shared storage solution for a gaming application that is hosted in the AWS Cloud. The company needs the ability to use SMB clients to access data. The solution must be fully managed.

Which AWS solution meets these requirements?

一家公司正在為 AWS 雲端中託管的遊戲應用程式設計共享儲存解決方案。該公司需要能夠使用 SMB 用戶端存取資料。該解決方案必須經過全面管理。

哪種 AWS 解決方案符合這些要求？

【選項】

A.Create an AWS DataSync task that shares the data as a mountable file system. Mount the file system to the application server.

B.Create an Amazon EC2 Windows instance. Install and configure a Windows file share role on the instance. Connect the application server to the file share.

C.Create an Amazon FSx for Windows File Server file system. Attach the file system to the origin server. Connect the application server to the file system.

D.Create an Amazon S3 bucket. Assign an IAM role to the application to grant access to the S3 bucket. Mount the S3 bucket to the application server.

【參考答案】C

【題號】question 306

【題目】

A company wants to run an in-memory database for a latency-sensitive application that runs on Amazon EC2 instances. The application processes more than 100,000 transactions each minute and requires high network throughput. A solutions architect needs to provide a cost-effective network design that minimizes data transfer charges.

Which solution meets these requirements?

一家公司希望為在 Amazon EC2 執行個體上執行的延遲敏感應用程式運行記憶體資料庫。該應用程式每分鐘處理超過100,000筆交易，並且需要高網路吞吐量。解決方案架構師需要提供經濟高效的網路設計，以最大限度地降低資料傳輸費用。

哪種解決方案滿足這些要求？

【選項】

A.Launch all EC2 instances in the same Availability Zone within the same AWS Region. Specify a placement group with cluster strategy when launching EC2 instances.

B.Launch all EC2 instances in different Availability Zones within the same AWS Region. Specify a placement group with partition strategy when launching EC2 instances.

C.Deploy an Auto Scaling group to launch EC2 instances in different Availability Zones based on a network utilization target.

D. Deploy an Auto Scaling group with a step scaling policy to launch EC2 instances in different Availability Zones.

【參考答案】A

【題號】question 307

【題目】

A company that primarily runs its application servers on premises has decided to migrate to AWS. The company wants to minimize its need to scale its Internet Small Computer Systems Interface (iSCSI) storage on premises. The company wants only its recently accessed data to remain stored locally.

Which AWS solution should the company use to meet these requirements?

一家主要在本地運行應用程式伺服器的公司決定遷移到 AWS。該公司希望最大限度地減少擴展其本地互聯網小型電腦系統介面 (iSCSI) 儲存的需求。該公司只希望將最近訪問的資料保留在本地。

公司應該使用哪種 AWS 解決方案來滿足這些要求？

【選項】

A. Amazon S3 File Gateway

B. AWS Storage Gateway Tape Gateway

C. AWS Storage Gateway Volume Gateway stored volumes

D. AWS Storage Gateway Volume Gateway cached volumes

【參考答案】D

【題號】question 308

【題目】

A company has multiple AWS accounts that use consolidated billing. The company runs several active high performance Amazon RDS for Oracle On-Demand DB instances for 90 days. The company's finance team has access to AWS Trusted Advisor in the consolidated billing account and all other AWS accounts.

The finance team needs to use the appropriate AWS account to access the Trusted Advisor check recommendations for RDS. The finance team must review the appropriate Trusted Advisor check to reduce RDS costs.

Which combination of steps should the finance team take to meet these requirements? (Choose two.)

一家公司擁有多個使用合併帳單的 AWS 帳戶。該公司為 Oracle 按需資料庫執行個體執行多個活動高效能 Amazon RDS 90 天。該公司的財務團隊可以存取合併計費帳戶和所有其他 AWS 帳戶中的 AWS Trusted Advisor。

財務團隊需要使用適當的 AWS 帳戶來存取 RDS 的 Trusted Advisor 檢查建議。財務團隊必須審查適當的 Trusted Advisor 檢查以降低 RDS 成本。

財務團隊應採取哪些步驟組合來滿足這些要求？（選兩個。）

【選項】

- A. Use the Trusted Advisor recommendations from the account where the RDS instances are running.
- B. Use the Trusted Advisor recommendations from the consolidated billing account to see all RDS instance checks at the same time.
- C. Review the Trusted Advisor check for Amazon RDS Reserved Instance Optimization.
- D. Review the Trusted Advisor check for Amazon RDS Idle DB Instances.
- E. Review the Trusted Advisor check for Amazon Redshift Reserved Node Optimization.

【參考答案】BD

【題號】question 309

【題目】

A solutions architect needs to optimize storage costs. The solutions architect must identify any Amazon S3 buckets that are no longer being accessed or are rarely accessed.

Which solution will accomplish this goal with the LEAST operational overhead?

解決方案架構師需要優化儲存成本。解決方案架構師必須識別不再被存取或很少存取的任何 Amazon S3 儲存桶。

哪種解決方案能夠以最少的營運開銷實現這一目標？

【選項】

A. Analyze bucket access patterns by using the S3 Storage Lens dashboard for advanced activity metrics.

B. Analyze bucket access patterns by using the S3 dashboard in the AWS Management Console.

C. Turn on the Amazon CloudWatch BucketSizeBytes metric for buckets. Analyze bucket access patterns by using the metrics data with Amazon Athena.

D. Turn on AWS CloudTrail for S3 object monitoring. Analyze bucket access patterns by using CloudTrail logs that are integrated with Amazon CloudWatch Logs.

【參考答案】A

【題號】question 310

【題目】

A company sells datasets to customers who do research in artificial intelligence and machine learning (AI/ML). The datasets are large, formatted files that are stored in an Amazon S3 bucket in the us-east-1 Region. The company hosts a web application that the customers use to purchase access to a given dataset. The web application is deployed on multiple Amazon EC2 instances behind an Application Load Balancer. After a purchase is made, customers receive an S3 signed URL that allows access to the files.

The customers are distributed across North America and Europe. The company wants to reduce the cost that is associated with data transfers and wants to maintain or improve performance.

What should a solutions architect do to meet these requirements?

一家公司向從事人工智慧和機器學習 (AI/ML) 研究的客戶出售資料集。資料集是大型格式化文件，儲存在 us-east-1 區域的 Amazon S3 儲存桶中。該公司託管一個 Web 應用程式，客戶可以使用該應用程式購買給定資料集的存取權。Web 應用程式部署在應用程式負載平衡器後面的多個 Amazon EC2 執行個體上。購買後，客戶會收到一個 S3 簽署的 URL，允許存取文件。

客戶遍布北美和歐洲。該公司希望降低與數據傳輸相關的成本，並希望維持或提高效能。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A. Configure S3 Transfer Acceleration on the existing S3 bucket. Direct customer requests to the S3 Transfer Acceleration endpoint. Continue to use S3 signed URLs for access control.

B. Deploy an Amazon CloudFront distribution with the existing S3 bucket as the origin. Direct customer requests to the CloudFront URL. Switch to CloudFront signed URLs for access control.

C. Set up a second S3 bucket in the eu-central-1 Region with S3 Cross-Region Replication between the buckets. Direct customer requests to the closest Region. Continue to use S3 signed URLs for access control.

D. Modify the web application to enable streaming of the datasets to end users. Configure the web application to read the data from the existing S3 bucket. Implement access control directly in the application.

【參考答案】B

【題號】question 311

【題目】

A company is using AWS to design a web application that will process insurance quotes. Users will request quotes from the application. Quotes must be separated by quote type, must be responded to within 24 hours, and must not get lost. The solution must maximize operational efficiency and must minimize maintenance.

Which solution meets these requirements?

一家公司正在使用 AWS 設計一個用於處理保險報價的 Web 應用程式。用戶將向應用程式請求報價。報價必須按報價類型分隔，必須在24小時內回復，且不得遺失。此解決方案必須最大限度地提高營運效率並最大限度地減少維護。

哪種解決方案滿足這些要求？

【選項】

A.Create multiple Amazon Kinesis data streams based on the quote type. Configure the web application to send messages to the proper data stream. Configure each backend group of application servers to use the Kinesis Client Library (KCL) to pool messages from its own data stream.

B.Create an AWS Lambda function and an Amazon Simple Notification Service (Amazon SNS) topic for each quote type. Subscribe the Lambda function to its associated SNS topic. Configure the application to publish requests for quotes to the appropriate SNS topic.

C.Create a single Amazon Simple Notification Service (Amazon SNS) topic. Subscribe Amazon Simple Queue Service (Amazon SQS) queues to the SNS topic. Configure SNS message filtering to publish messages to the proper SQS queue based on the quote type. Configure each backend application server to use its own SQS queue.

D.Create multiple Amazon Kinesis Data Firehose delivery streams based on the quote type to deliver data streams to an Amazon OpenSearch Service cluster. Configure the application to send messages to the proper delivery stream. Configure each backend group of application servers to search for the messages from OpenSearch Service and process them accordingly.

【參考答案】C

【題號】question 312

【題目】

A company has an application that runs on several Amazon EC2 instances. Each EC2 instance has multiple Amazon Elastic Block Store (Amazon EBS) data volumes attached to it. The application's EC2 instance configuration and data need to be backed up nightly. The application also needs to be recoverable in a different AWS Region.

Which solution will meet these requirements in the MOST operationally efficient way?

一家公司有一個在多個 Amazon EC2 執行個體上執行的應用程式。每個 EC2 執行個體都附加多個 Amazon Elastic Block Store (Amazon EBS) 資料磁碟區。應用程式的 EC2 執行個體配置和資料需要每晚備份。該應用程式還需要在不同的 AWS 區域中可恢復。

哪一種解決方案能夠以最高效的方式滿足這些要求？

【選項】

A. Write an AWS Lambda function that schedules nightly snapshots of the application's EBS volumes and copies the snapshots to a different Region.

B. Create a backup plan by using AWS Backup to perform nightly backups. Copy the backups to another Region. Add the application's EC2 instances as resources.

C. Create a backup plan by using AWS Backup to perform nightly backups. Copy the backups to another Region. Add the application's EBS volumes as resources.

D. Write an AWS Lambda function that schedules nightly snapshots of the application's EBS volumes and copies the snapshots to a different Availability Zone.

【參考答案】B

【題號】question 313

【題目】

A company is building a mobile app on AWS. The company wants to expand its reach to millions of users. The company needs to build a platform so that authorized users can watch the company's content on their mobile devices.

What should a solutions architect recommend to meet these requirements?

一家公司正在 AWS 上建立行動應用程式。該公司希望將其覆蓋範圍擴大到數百萬用戶。公司需要建立一個平台，以便授權使用者可以在行動裝置上觀看公司的內容。

解決方案架構師應該建議什麼來滿足這些要求？

【選項】

- A.Publish content to a public Amazon S3 bucket. Use AWS Key Management Service (AWS KMS) keys to stream content.
- B.Set up IPsec VPN between the mobile app and the AWS environment to stream content.
- C.Use Amazon CloudFront. Provide signed URLs to stream content.
- D.Set up AWS Client VPN between the mobile app and the AWS environment to stream content.

【參考答案】C

【題號】question 314

【題目】

A company has an on-premises MySQL database used by the global sales team with infrequent access patterns. The sales team requires the database to have minimal downtime. A database administrator wants to migrate this database to AWS without selecting a particular instance type in anticipation of more users in the future.

Which service should a solutions architect recommend?

一家公司擁有一個本地 MySQL 資料庫，供全球銷售團隊使用，且存取模式不頻繁。銷售團隊要求資料庫的停機時間最少。資料庫管理員希望將此資料庫遷移到 AWS，而不選擇特定執行個體類型，以應對未來更多使用者的需求。

解決方案架構師應該推薦哪一種服務？

【選項】

- A.Amazon Aurora MySQL
- B.Amazon Aurora Serverless for MySQL
- C.Amazon Redshift Spectrum
- D.Amazon RDS for MySQL

【參考答案】B

【題號】question 315

【題目】

A company experienced a breach that affected several applications in its on-premises data center. The attacker took advantage of vulnerabilities in the custom applications that were running on the servers. The company is now migrating its applications to run on Amazon EC2 instances. The company wants to implement a solution that actively scans for vulnerabilities on the EC2 instances and sends a report that details the findings.

Which solution will meet these requirements?

一家公司遭遇了一次漏洞，影響了其本地資料中心的多個應用程式。攻擊者利用了伺服器上運行的自訂應用程式中的漏洞。該公司目前正在將其應用程式遷移到在 Amazon EC2 執行個體上運行。該公司希望實施一種解決方案，主動掃描 EC2 執行個體上的漏洞並發送詳細說明發現結果的報告。

哪種解決方案可以滿足這些要求？

【選項】

- A.Deploy AWS Shield to scan the EC2 instances for vulnerabilities. Create an AWS Lambda function to log any findings to AWS CloudTrail.
- B.Deploy Amazon Macie and AWS Lambda functions to scan the EC2 instances for vulnerabilities. Log any findings to AWS CloudTrail.
- C.Turn on Amazon GuardDuty. Deploy the GuardDuty agents to the EC2 instances. Configure an AWS Lambda function to automate the generation and distribution of reports that detail the findings.

D.Turn on Amazon Inspector. Deploy the Amazon Inspector agent to the EC2 instances. Configure an AWS Lambda function to automate the generation and distribution of reports that detail the findings.

【參考答案】D

【題號】question 316

【題目】

A company uses an Amazon EC2 instance to run a script to poll for and process messages in an Amazon Simple Queue Service (Amazon SQS) queue. The company wants to reduce operational costs while maintaining its ability to process a growing number of messages that are added to the queue.

What should a solutions architect recommend to meet these requirements?

一家公司使用 Amazon EC2 執行個體執行腳本來輪詢並處理 Amazon Simple Queue Service (Amazon SQS) 佇列中的消息。該公司希望降低營運成本，同時保持處理越來越多添加到隊列中的消息的能力。

解決方案架構師應該建議什麼來滿足這些要求？

【選項】

A.Increase the size of the EC2 instance to process messages faster.

B.Use Amazon EventBridge to turn off the EC2 instance when the instance is underutilized.

C.Migrate the script on the EC2 instance to an AWS Lambda function with the appropriate runtime.

D.Use AWS Systems Manager Run Command to run the script on demand.

【參考答案】C

【題號】question 317

【題目】

A company uses a legacy application to produce data in CSV format. The legacy application stores the output data in Amazon S3. The company is deploying a new commercial off-the-shelf (COTS) application that can perform complex SQL queries to analyze data that is stored in Amazon Redshift and Amazon S3 only. However, the COTS application cannot process the .csv files that the legacy application produces.

The company cannot update the legacy application to produce data in another format. The company needs to implement a solution so that the COTS application can use the data that the legacy application produces.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司使用舊應用程式產生 CSV 格式的資料。舊應用程式將輸出資料儲存在 Amazon S3 中。該公司正在部署一個新的商業現成 (COTS) 應用程式，該應用程式可以執行複雜的 SQL 查詢來分析僅儲存在 Amazon Redshift 和 Amazon S3 中的資料。但是，COTS 應用程式無法處理舊應用程式產生的 .csv 檔案。

該公司無法更新舊應用程式以產生其他格式的數據。該公司需要實施一個解決方案，以便 COTS 應用程式可以使用舊應用程式產生的資料。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A.Create an AWS Glue extract, transform, and load (ETL) job that runs on a schedule.

Configure the ETL job to process the .csv files and store the processed data in Amazon Redshift.

B.Develop a Python script that runs on Amazon EC2 instances to convert the .csv files to .sql files. Invoke the Python script on a cron schedule to store the output files in Amazon S3.

C.Create an AWS Lambda function and an Amazon DynamoDB table. Use an S3 event to invoke the Lambda function. Configure the Lambda function to perform an extract, transform, and load (ETL) job to process the .csv files and store the processed data in the DynamoDB table.

D.Use Amazon EventBridge to launch an Amazon EMR cluster on a weekly schedule. Configure the EMR cluster to perform an extract, transform, and load (ETL) job to process the .csv files and store the processed data in an Amazon Redshift table.

【參考答案】A

【題號】question 318

【題目】

A company recently migrated its entire IT environment to the AWS Cloud. The company discovers that users are provisioning oversized Amazon EC2 instances and modifying security group rules without using the appropriate change control process. A solutions architect must devise a strategy to track and audit these inventory and configuration changes.

Which actions should the solutions architect take to meet these requirements? (Choose two.)

一家公司最近將整個 IT 環境遷移到 AWS 雲端。該公司發現用戶在未使用適當的變更控制流程的情況下配置超大的 Amazon EC2 執行個體並修改安全性群組規則。解決方案架構師必須制定策略來追蹤和審核這些庫存和配置變更。

解決方案架構師應該採取哪些行動來滿足這些要求？（選兩個。）

【選項】

- A.Enable AWS CloudTrail and use it for auditing.
- B.Use data lifecycle policies for the Amazon EC2 instances.
- C.Enable AWS Trusted Advisor and reference the security dashboard.
- D.Enable AWS Config and create rules for auditing and compliance purposes.
- E.Restore previous resource configurations with an AWS CloudFormation template.

【參考答案】AD

【題號】question 319

【題目】

A company has hundreds of Amazon EC2 Linux-based instances in the AWS Cloud. Systems administrators have used shared SSH keys to manage the instances. After a recent audit, the company's security team is mandating the removal of all shared keys. A solutions architect must design a solution that provides secure access to the EC2 instances.

Which solution will meet this requirement with the LEAST amount of administrative overhead?

一家公司在 AWS 雲端中擁有數百個基於 Linux 的 Amazon EC2 執行個體。系統管理員使用共用 SSH 金鑰來管理執行個體。經過最近的審計後，該公司的安全團隊要求刪除所有共享金鑰。解決方案架構師必須設計一個能夠提供對 EC2 執行個體的安全存取的解決方案。

哪種解決方案能夠以最少的管理開銷滿足此要求？

【選項】

- A. Use AWS Systems Manager Session Manager to connect to the EC2 instances.
- B. Use AWS Security Token Service (AWS STS) to generate one-time SSH keys on demand.
- C. Allow shared SSH access to a set of bastion instances. Configure all other instances to allow only SSH access from the bastion instances.
- D. Use an Amazon Cognito custom authorizer to authenticate users. Invoke an AWS Lambda function to generate a temporary SSH key.

【參考答案】A

【題號】question 320

【題目】

A company is using a fleet of Amazon EC2 instances to ingest data from on-premises data sources. The data is in JSON format and ingestion rates can be as high as 1 MB/s.

When an EC2 instance is rebooted, the data in-flight is lost. The company's data science team wants to query ingested data in near-real time.

Which solution provides near-real-time data querying that is scalable with minimal data loss?

一家公司正在使用一組 Amazon EC2 執行個體從本地資料來源提取資料。資料採用 JSON 格式，攝取速率可高達 1 MB/s。當 EC2 執行個體重新啟動時，傳輸中的資料將會遺失。該公司的數據科學團隊希望近乎即時地查詢所取得的數據。

哪種解決方案提供可擴展且資料遺失最少的近即時資料查詢？

【選項】

A.Publish data to Amazon Kinesis Data Streams, Use Kinesis Data Analytics to query the data.

B.Publish data to Amazon Kinesis Data Firehose with Amazon Redshift as the destination. Use Amazon Redshift to query the data.

C.Store ingested data in an EC2 instance store. Publish data to Amazon Kinesis Data Firehose with Amazon S3 as the destination. Use Amazon Athena to query the data.

D.Store ingested data in an Amazon Elastic Block Store (Amazon EBS) volume. Publish data to Amazon ElastiCache for Redis. Subscribe to the Redis channel to query the data.

【參考答案】A

【題號】question 321

【題目】

What should a solutions architect do to ensure that all objects uploaded to an Amazon S3 bucket are encrypted?

解決方案架構師應該如何確保上傳到 Amazon S3 儲存桶的所有物件都經過加密？

【選項】

A.Update the bucket policy to deny if the PutObject does not have an s3:x-amz-acl header set.

B.Update the bucket policy to deny if the PutObject does not have an s3:x-amz-acl header set to private.

C.Update the bucket policy to deny if the PutObject does not have an aws:SecureTransport header set to true.

D.Update the bucket policy to deny if the PutObject does not have an x-amz-server-side-encryption header set.

【參考答案】D

【題號】question 322

【題目】

A solutions architect is designing a multi-tier application for a company. The application's users upload images from a mobile device. The application generates a thumbnail of each image and returns a message to the user to confirm that the image was uploaded successfully.

The thumbnail generation can take up to 60 seconds, but the company wants to provide a faster response time to its users to notify them that the original image was received. The solutions architect must design the application to asynchronously dispatch requests to the different application tiers.

What should the solutions architect do to meet these requirements?

解決方案架構師正在為公司設計多層應用程式。該應用程式的用戶從行動裝置上傳圖像。該應用程式產生每個圖像的縮圖並向用戶返回一條訊息以確認圖像已成功上傳。

縮圖生成可能需要長達 60 秒的時間，但該公司希望向用戶提供更快的回應時間，以通知他們已收到原始圖像。解決方案架構師必須設計應用程式以將請求非同步分派到不同的應用程式層。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A. Write a custom AWS Lambda function to generate the thumbnail and alert the user. Use the image upload process as an event source to invoke the Lambda function.

B. Create an AWS Step Functions workflow. Configure Step Functions to handle the orchestration between the application tiers and alert the user when thumbnail generation is complete.

C. Create an Amazon Simple Queue Service (Amazon SQS) message queue. As images are uploaded, place a message on the SQS queue for thumbnail generation. Alert the user through an application message that the image was received.

D. Create Amazon Simple Notification Service (Amazon SNS) notification topics and subscriptions. Use one subscription with the application to generate the thumbnail after the image upload is complete. Use a second subscription to message the user's mobile app by way of a push notification after thumbnail generation is complete.

【參考答案】C

【題號】question 323

【題目】

A company's facility has badge readers at every entrance throughout the building. When badges are scanned, the readers send a message over HTTPS to indicate who attempted to access that particular entrance.

A solutions architect must design a system to process these messages from the sensors. The solution must be highly available, and the results must be made available for the company's security team to analyze.

Which system architecture should the solutions architect recommend?

一家公司的設施在整個建築的每個入口處都有徽章閱讀器。掃描徽章後，閱讀器會透過 HTTPS 發送一條訊息，指示誰試圖存取該特定入口。

解決方案架構師必須設計一個系統來處理來自感測器的這些訊息。此解決方案必須具有高可用性，且結果必須可供公司的安全團隊進行分析。

解決方案架構師應該推薦哪種系統架構？

【選項】

A.Launch an Amazon EC2 instance to serve as the HTTPS endpoint and to process the messages. Configure the EC2 instance to save the results to an Amazon S3 bucket.

B.Create an HTTPS endpoint in Amazon API Gateway. Configure the API Gateway endpoint to invoke an AWS Lambda function to process the messages and save the results to an Amazon DynamoDB table.

C.Use Amazon Route 53 to direct incoming sensor messages to an AWS Lambda function. Configure the Lambda function to process the messages and save the results to an Amazon DynamoDB table.

D.Create a gateway VPC endpoint for Amazon S3. Configure a Site-to-Site VPN connection from the facility network to the VPC so that sensor data can be written directly to an S3 bucket by way of the VPC endpoint.

【參考答案】B

【題號】question 324

【題目】

A company wants to implement a disaster recovery plan for its primary on-premises file storage volume. The file storage volume is mounted from an Internet Small Computer Systems Interface (iSCSI) device on a local storage server. The file storage volume holds hundreds of terabytes (TB) of data.

The company wants to ensure that end users retain immediate access to all file types from the on-premises systems without experiencing latency.

Which solution will meet these requirements with the LEAST amount of change to the company's existing infrastructure?

一家公司希望為其主要本地文件儲存卷實施災難復原計畫。檔案儲存磁碟區是從本機儲存伺服器上的 Internet 小型電腦系統介面 (iSCSI) 裝置安裝的。檔案儲存磁碟區可容納數百太位元組 (TB) 的資料。

該公司希望確保最終用戶能夠從本機系統立即存取所有文件類型，而不會遇到延遲。

哪種解決方案能夠滿足這些要求，並且對公司現有基礎設施的改變最少？

【選項】

A.Provision an Amazon S3 File Gateway as a virtual machine (VM) that is hosted on premises. Set the local cache to 10 TB. Modify existing applications to access the files through the NFS protocol. To recover from a disaster, provision an Amazon EC2 instance and mount the S3 bucket that contains the files.

B.Provision an AWS Storage Gateway tape gateway. Use a data backup solution to back up all existing data to a virtual tape library. Configure the data backup solution to run nightly after the initial backup is complete. To recover from a disaster, provision an Amazon EC2 instance and restore the data to an Amazon Elastic Block Store (Amazon EBS) volume from the volumes in the virtual tape library.

C.Provision an AWS Storage Gateway Volume Gateway cached volume. Set the local cache to 10 TB. Mount the Volume Gateway cached volume to the existing file server by using iSCSI, and copy all files to the storage volume. Configure scheduled snapshots of the storage volume. To recover from a disaster, restore a snapshot to an Amazon Elastic Block Store (Amazon EBS) volume and attach the EBS volume to an Amazon EC2 instance.

D.Provision an AWS Storage Gateway Volume Gateway stored volume with the same amount of disk space as the existing file storage volume. Mount the Volume Gateway stored volume to the existing file server by using iSCSI, and copy all files to the storage volume. Configure scheduled snapshots of the storage volume. To recover from a disaster, restore a snapshot to an Amazon Elastic Block Store (Amazon EBS) volume and attach the EBS volume to an Amazon EC2 instance.

【參考答案】D

【題號】question 325

【題目】

A company is hosting a web application from an Amazon S3 bucket. The application uses Amazon Cognito as an identity provider to authenticate users and return a JSON Web Token (JWT) that provides access to protected resources that are stored in another S3 bucket.

Upon deployment of the application, users report errors and are unable to access the protected content. A solutions architect must resolve this issue by providing proper permissions so that users can access the protected content.

Which solution meets these requirements?

一家公司正在託管 Amazon S3 儲存桶中的 Web 應用程式。該應用程式使用 Amazon Cognito 作為身分提供者來對使用者進行身份驗證並傳回 JSON Web 令牌 (JWT)，該令牌提供對儲存在另一個 S3 儲存桶中的受保護資源的存取權。

部署應用程式後，使用者報告錯誤並且無法存取受保護的內容。解決方案架構師必須透過提供適當的權限來解決此問題，以便使用者可以存取受保護的內容。

哪種解決方案滿足這些要求？

【選項】

- A. Update the Amazon Cognito identity pool to assume the proper IAM role for access to the protected content.
- B. Update the S3 ACL to allow the application to access the protected content.
- C. Redeploy the application to Amazon S3 to prevent eventually consistent reads in the S3 bucket from affecting the ability of users to access the protected content.
- D. Update the Amazon Cognito pool to use custom attribute mappings within the identity pool and grant users the proper permissions to access the protected content.

【參考答案】A

【題號】question 326

【題目】

An image hosting company uploads its large assets to Amazon S3 Standard buckets. The company uses multipart upload in parallel by using S3 APIs and overwrites if the same object is uploaded again. For the first 30 days after upload, the objects will be accessed frequently. The objects will be used less frequently after 30 days, but the access patterns for each object will be inconsistent. The company must optimize its S3 storage costs while maintaining high availability and resiliency of stored assets.

Which combination of actions should a solutions architect recommend to meet these requirements? (Choose two.)

一家圖像託管公司將其大量資產上傳到 Amazon S3 標準儲存桶。該公司透過使用 S3 API 並行進行分段上傳，如果再次上傳相同的對象，則會進行覆寫。上傳後的前 30 天內，物件將被頻繁存取。30 天后，物件的使用頻率將降低，但每個物件的存取模式將不一致。該公司必須優化其 S3 儲存成本，同時保持儲存資產的高可用性和彈性。

解決方案架構師應該推薦哪種操作組合來滿足這些要求？（選兩個。）

【選項】

- A. Move assets to S3 Intelligent-Tiering after 30 days.
- B. Configure an S3 Lifecycle policy to clean up incomplete multipart uploads.
- C. Configure an S3 Lifecycle policy to clean up expired object delete markers.
- D. Move assets to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days.
- E. Move assets to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 30 days.

【參考答案】AB

【題號】question 327

【題目】

A solutions architect must secure a VPC network that hosts Amazon EC2 instances. The EC2 instances contain highly sensitive data and run in a private subnet. According to company policy, the EC2 instances that run in the VPC can access only approved third-party software repositories on the internet for software product updates that use the third party's URL. Other internet traffic must be blocked.

Which solution meets these requirements?

解決方案架構師必須保護託管 Amazon EC2 執行個體的 VPC 網路。EC2 執行個體包含高度敏感的資料並在私有子網路中運作。根據公司政策，在 VPC 中運行的 EC2 執行個體只能存取互聯網上經批准的第三方軟體儲存庫，以使用第三方 URL 進行軟體產品更新。必須阻止其他網路流量。

哪種解決方案滿足這些要求？

【選項】

A.Update the route table for the private subnet to route the outbound traffic to an AWS Network Firewall firewall. Configure domain list rule groups.

B.Set up an AWS WAF web ACL. Create a custom set of rules that filter traffic requests based on source and destination IP address range sets.

C.Implement strict inbound security group rules. Configure an outbound rule that allows traffic only to the authorized software repositories on the internet by specifying the URLs.

D.Configure an Application Load Balancer (ALB) in front of the EC2 instances. Direct all outbound traffic to the ALB. Use a URL-based rule listener in the ALB's target group for outbound access to the internet.

【參考答案】A

【題號】question 328

【題目】

A company is hosting a three-tier ecommerce application in the AWS Cloud. The company hosts the website on Amazon S3 and integrates the website with an API that handles sales requests. The company hosts the API on three Amazon EC2 instances behind an Application Load Balancer (ALB). The API consists of static and dynamic front-end content along with backend workers that process sales requests asynchronously.

The company is expecting a significant and sudden increase in the number of sales requests during events for the launch of new products.

What should a solutions architect recommend to ensure that all the requests are processed successfully?

一家公司在 AWS 雲端中託管一個三層電子商務應用程式。該公司在 Amazon S3 上託管該網站，並將該網站與處理銷售請求的 API 整合。該公司在應用程式負載平衡器 (ALB) 後面的三個 Amazon EC2 執行個體上託管 API。此 API 由靜態和動態前端內容以及非同步處理銷售請求的後端工作人員組成。

該公司預計，在新產品發布活動期間，銷售請求數量將大幅突然增加。

解決方案架構師應該建議什麼來確保成功處理所有請求？

【選項】

A.Add an Amazon CloudFront distribution for the dynamic content. Increase the number of EC2 instances to handle the increase in traffic.

B.Add an Amazon CloudFront distribution for the static content. Place the EC2 instances in an Auto Scaling group to launch new instances based on network traffic.

C.Add an Amazon CloudFront distribution for the dynamic content. Add an Amazon ElastiCache instance in front of the ALB to reduce traffic for the API to handle.

D.Add an Amazon CloudFront distribution for the static content. Add an Amazon Simple Queue Service (Amazon SQS) queue to receive requests from the website for later processing by the EC2 instances.

【參考答案】D

【題號】question 329

【題目】

A security audit reveals that Amazon EC2 instances are not being patched regularly. A solutions architect needs to provide a solution that will run regular security scans across a large fleet of EC2 instances. The solution should also patch the EC2 instances on a regular schedule and provide a report of each instance's patch status.

Which solution will meet these requirements?

安全性稽核顯示 Amazon EC2 執行個體沒有定期修補。解決方案架構師需要提供一種解決方案，能夠對大量 EC2 執行個體執行定期安全掃描。此解決方案還應定期修補 EC2 實例，並提供每個執行個體的修補狀態報告。

哪種解決方案可以滿足這些要求？

【選項】

A.Set up Amazon Macie to scan the EC2 instances for software vulnerabilities. Set up a cron job on each EC2 instance to patch the instance on a regular schedule.

B.Turn on Amazon GuardDuty in the account. Configure GuardDuty to scan the EC2 instances for software vulnerabilities. Set up AWS Systems Manager Session Manager to patch the EC2 instances on a regular schedule.

C.Set up Amazon Detective to scan the EC2 instances for software vulnerabilities. Set up an Amazon EventBridge scheduled rule to patch the EC2 instances on a regular schedule.

D.Turn on Amazon Inspector in the account. Configure Amazon Inspector to scan the EC2 instances for software vulnerabilities. Set up AWS Systems Manager Patch Manager to patch the EC2 instances on a regular schedule.

【參考答案】D

【題號】question 330

【題目】

A company is planning to store data on Amazon RDS DB instances. The company must encrypt the data at rest.

What should a solutions architect do to meet this requirement?

一家公司計劃在 Amazon RDS 資料庫執行個體上儲存資料。公司必須對靜態資料進行加密。
解決方案架構師應該怎麼做才能滿足這項要求？

【選項】

A.Create a key in AWS Key Management Service (AWS KMS). Enable encryption for the DB instances.

B.Create an encryption key. Store the key in AWS Secrets Manager. Use the key to encrypt the DB instances.

C.Generate a certificate in AWS Certificate Manager (ACM). Enable SSL/TLS on the DB instances by using the certificate.

D.Generate a certificate in AWS Identity and Access Management (IAM). Enable SSL/TLS on the DB instances by using the certificate.

【參考答案】A

【題號】question 331

【題目】

A company must migrate 20 TB of data from a data center to the AWS Cloud within 30 days. The company's network bandwidth is limited to 15 Mbps and cannot exceed 70% utilization.

What should a solutions architect do to meet these requirements?

公司必須在 30 天內將 20 TB 資料從資料中心遷移到 AWS 雲端。該公司的網路頻寬限制為 15 Mbps, 利用率不能超過 70%。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

- A.Use AWS Snowball.
- B.Use AWS DataSync.
- C.Use a secure VPN connection.
- D.Use Amazon S3 Transfer Acceleration.

【參考答案】A

【題號】question 332

【題目】

A company needs to provide its employees with secure access to confidential and sensitive files. The company wants to ensure that the files can be accessed only by authorized users. The files must be downloaded securely to the employees' devices.

The files are stored in an on-premises Windows file server. However, due to an increase in remote usage, the file server is running out of capacity.

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Which solution will meet these requirements?

公司需要為其員工提供對機密和敏感文件的安全存取。該公司希望確保只有授權使用者才能存取這些文件。文件必須安全地下載到員工的裝置上。

這些檔案儲存在本機 Windows 檔案伺服器中。然而，由於遠端使用量的增加，檔案伺服器的容量即將耗盡。

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哪種解決方案可以滿足這些要求？

【選項】

A.Migrate the file server to an Amazon EC2 instance in a public subnet. Configure the security group to limit inbound traffic to the employees' IP addresses.

B.Migrate the files to an Amazon FSx for Windows File Server file system. Integrate the Amazon FSx file system with the on-premises Active Directory. Configure AWS Client VPN.

C.Migrate the files to Amazon S3, and create a private VPC endpoint. Create a signed URL to allow download.

D.Migrate the files to Amazon S3, and create a public VPC endpoint. Allow employees to sign on with AWS IAM Identity Center (AWS Single Sign-On).

【參考答案】B

【題號】question 333

【題目】

A company's application runs on Amazon EC2 instances behind an Application Load Balancer (ALB). The instances run in an Amazon EC2 Auto Scaling group across multiple Availability Zones. On the first day of every month at midnight, the application becomes much slower when the month-end financial calculation batch runs. This causes the CPU utilization of the EC2 instances to immediately peak to 100%, which disrupts the application.

What should a solutions architect recommend to ensure the application is able to handle the workload and avoid downtime?

該公司的應用程式在應用程式負載平衡器 (ALB) 後面的 Amazon EC2 執行個體上執行。這些執行個體在跨多個可用區的 Amazon EC2 Auto Scaling 群組中運作。在每個月的第一天午夜，當月末財務計算批次運行時，應用程式會變得更慢。這會導致 EC2 執行個體的 CPU 使用率立即達到 100% 峰值，從而中斷應用程式。

解決方案架構師應該建議什麼來確保應用程式能夠處理工作負載並避免停機？

【選項】

- A. Configure an Amazon CloudFront distribution in front of the ALB.
- B. Configure an EC2 Auto Scaling simple scaling policy based on CPU utilization.
- C. Configure an EC2 Auto Scaling scheduled scaling policy based on the monthly schedule.
- D. Configure Amazon ElastiCache to remove some of the workload from the EC2 instances.

【參考答案】C

【題號】question 334

【題目】

A company wants to give a customer the ability to use on-premises Microsoft Active Directory to download files that are stored in Amazon S3. The customer's application uses an SFTP client to download the files.

Which solution will meet these requirements with the LEAST operational overhead and no changes to the customer's application?

一家公司希望讓客戶能夠使用本機 Microsoft Active Directory 下載儲存在 Amazon S3 中的檔案。客戶的應用程式使用 SFTP 用戶端下載檔案。

哪種解決方案能夠以最少的營運開銷滿足這些要求，並且無需更改客戶的應用程式？

【選項】

A.Set up AWS Transfer Family with SFTP for Amazon S3. Configure integrated Active Directory authentication.

B.Set up AWS Database Migration Service (AWS DMS) to synchronize the on-premises client with Amazon S3. Configure integrated Active Directory authentication.

C.Set up AWS DataSync to synchronize between the on-premises location and the S3 location by using AWS IAM Identity Center (AWS Single Sign-On).

D.Set up a Windows Amazon EC2 instance with SFTP to connect the on-premises client with Amazon S3. Integrate AWS Identity and Access Management (IAM).

【參考答案】A

【題號】question 335

【題目】

A company is experiencing sudden increases in demand. The company needs to provision large Amazon EC2 instances from an Amazon Machine Image (AMI). The instances will run in an Auto Scaling group. The company needs a solution that provides minimum initialization latency to meet the demand.

Which solution meets these requirements?

一家公司的需求突然增加。該公司需要從 Amazon 系統映像 (AMI) 預置大型 Amazon EC2 執行個體。這些執行個體將會在 Auto Scaling 群組中執行。該公司需要一種能夠提供最小初始化延遲的解決方案來滿足需求。

哪種解決方案滿足這些要求？

【選項】

- A. Use the `aws ec2 register-image` command to create an AMI from a snapshot. Use AWS Step Functions to replace the AMI in the Auto Scaling group.
- B. Enable Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot. Provision an AMI by using the snapshot. Replace the AMI in the Auto Scaling group with the new AMI.
- C. Enable AMI creation and define lifecycle rules in Amazon Data Lifecycle Manager (Amazon DLM). Create an AWS Lambda function that modifies the AMI in the Auto Scaling group.
- D. Use Amazon EventBridge to invoke AWS Backup lifecycle policies that provision AMIs. Configure Auto Scaling group capacity limits as an event source in EventBridge.

【參考答案】B

【題號】question 336

【題目】

A company hosts a multi-tier web application that uses an Amazon Aurora MySQL DB cluster for storage. The application tier is hosted on Amazon EC2 instances. The company's IT security guidelines mandate that the database credentials be encrypted and rotated every 14 days.

What should a solutions architect do to meet this requirement with the LEAST operational effort?

一家公司託管一個多層 Web 應用程式，該應用程式使用 Amazon Aurora MySQL 資料庫叢集進行儲存。應用程式層託管在 Amazon EC2 執行個體上。該公司的 IT 安全準則要求對資料庫憑證進行加密並每 14 天輪換一次。

解決方案架構師應該如何做才能以最少的操作工作來滿足此要求？

【選項】

A.Create a new AWS Key Management Service (AWS KMS) encryption key. Use AWS Secrets Manager to create a new secret that uses the KMS key with the appropriate credentials. Associate the secret with the Aurora DB cluster. Configure a custom rotation period of 14 days.

B.Create two parameters in AWS Systems Manager Parameter Store: one for the user name as a string parameter and one that uses the SecureString type for the password. Select AWS Key Management Service (AWS KMS) encryption for the password parameter, and load these parameters in the application tier. Implement an AWS Lambda function that rotates the password every 14 days.

C.Store a file that contains the credentials in an AWS Key Management Service (AWS KMS) encrypted Amazon Elastic File System (Amazon EFS) file system. Mount the EFS file system in all EC2 instances of the application tier. Restrict the access to the file on the file system so that the application can read the file and that only super users can modify the file. Implement an AWS Lambda function that rotates the key in Aurora every 14 days and writes new credentials into the file.

D.Store a file that contains the credentials in an AWS Key Management Service (AWS KMS) encrypted Amazon S3 bucket that the application uses to load the credentials. Download the file to the application regularly to ensure that the correct credentials are used. Implement an AWS Lambda function that rotates the Aurora credentials every 14 days and uploads these credentials to the file in the S3 bucket.

【參考答案】A

【題號】question 337

【題目】

A company has deployed a web application on AWS. The company hosts the backend database on Amazon RDS for MySQL with a primary DB instance and five read replicas to support scaling needs. The read replicas must lag no more than 1 second behind the primary DB instance. The database routinely runs scheduled stored procedures.

As traffic on the website increases, the replicas experience additional lag during periods of peak load. A solutions architect must reduce the replication lag as much as possible. The solutions architect must minimize changes to the application code and must minimize ongoing operational overhead.

Which solution will meet these requirements?

一家公司在 AWS 上部署了一個 Web 應用程式。該公司在 Amazon RDS for MySQL 上託管後端資料庫，並配備一個主資料庫執行個體和五個唯讀副本來支援擴充需求。只讀副本落後主資料庫執行個體的時間不得超過 1 秒。資料庫定期運行預定的預存程序。

隨著網站流量的增加，副本在尖峰負載期間會出現額外的延遲。解決方案架構師必須盡可能減少複製延遲。解決方案架構師必須最大限度地減少對應用程式程式碼的更改，並且必須最大限度地減少持續的營運開銷。

哪種解決方案可以滿足這些要求？

【選項】

A.Migrate the database to Amazon Aurora MySQL. Replace the read replicas with Aurora Replicas, and configure Aurora Auto Scaling. Replace the stored procedures with Aurora MySQL native functions.

B.Deploy an Amazon ElastiCache for Redis cluster in front of the database. Modify the application to check the cache before the application queries the database. Replace the stored procedures with AWS Lambda functions.

C.Migrate the database to a MySQL database that runs on Amazon EC2 instances. Choose large, compute optimized EC2 instances for all replica nodes. Maintain the stored procedures on the EC2 instances.

D.Migrate the database to Amazon DynamoDB. Provision a large number of read capacity units (RCUs) to support the required throughput, and configure on-demand capacity scaling. Replace the stored procedures with DynamoDB streams.

【參考答案】A

【題號】question 338

【題目】

A solutions architect must create a disaster recovery (DR) plan for a high-volume software as a service (SaaS) platform. All data for the platform is stored in an Amazon Aurora MySQL DB cluster.

The DR plan must replicate data to a secondary AWS Region.

Which solution will meet these requirements MOST cost-effectively?

解決方案架構師必須為大容量軟體即服務 (SaaS) 平台建立災難復原 (DR) 計畫。該平台的所有資料都儲存在 Amazon Aurora MySQL 資料庫叢集中。

DR 計畫必須將資料複製到輔助 AWS 區域。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A. Use MySQL binary log replication to an Aurora cluster in the secondary Region. Provision one DB instance for the Aurora cluster in the secondary Region.

B. Set up an Aurora global database for the DB cluster. When setup is complete, remove the DB instance from the secondary Region.

C. Use AWS Database Migration Service (AWS DMS) to continuously replicate data to an Aurora cluster in the secondary Region. Remove the DB instance from the secondary Region.

D. Set up an Aurora global database for the DB cluster. Specify a minimum of one DB instance in the secondary Region.

【參考答案】D

【題號】question 339

【題目】

A company has a custom application with embedded credentials that retrieves information from an Amazon RDS MySQL DB instance. Management says the application must be made more secure with the least amount of programming effort.

What should a solutions architect do to meet these requirements?

一家公司有一個帶有嵌入式憑證的自訂應用程式，用於從 Amazon RDS MySQL 資料庫實例檢索資訊。管理層表示，必須以最少的程式設計工作使應用程式更加安全。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A. Use AWS Key Management Service (AWS KMS) to create keys. Configure the application to load the database credentials from AWS KMS. Enable automatic key rotation.

B. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Create an AWS Lambda function that rotates the credentials in Secret Manager.

C. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Set up a credentials rotation schedule for the application user in the RDS for MySQL database using Secrets Manager.

D. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Systems Manager Parameter Store. Configure the application to load the database credentials from Parameter Store. Set up a credentials rotation schedule for the application user in the RDS for MySQL database using Parameter Store.

【參考答案】C

【題號】question 340

【題目】

A media company hosts its website on AWS. The website application's architecture includes a fleet of Amazon EC2 instances behind an Application Load Balancer (ALB) and a database that is hosted on Amazon Aurora. The company's cybersecurity team reports that the application is vulnerable to SQL injection.

How should the company resolve this issue?

一家媒體公司在 AWS 上託管其網站。此網站應用程式的架構包括應用程式負載平衡器 (ALB) 後面的一組 Amazon EC2 執行個體和託管在 Amazon Aurora 上的資料庫。該公司的網路安全團隊報告稱，該應用程式容易受到 SQL 注入攻擊。

公司該如何解決這個問題？

【選項】

- A. Use AWS WAF in front of the ALB. Associate the appropriate web ACLs with AWS WAF.
- B. Create an ALB listener rule to reply to SQL injections with a fixed response.
- C. Subscribe to AWS Shield Advanced to block all SQL injection attempts automatically.
- D. Set up Amazon Inspector to block all SQL injection attempts automatically.

【參考答案】A

【題號】question 341

【題目】

A company has an Amazon S3 data lake that is governed by AWS Lake Formation. The company wants to create a visualization in Amazon QuickSight by joining the data in the data lake with operational data that is stored in an Amazon Aurora MySQL database. The company wants to enforce column-level authorization so that the company's marketing team can access only a subset of columns in the database.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司擁有一個由 AWS Lake Formation 管理的 Amazon S3 資料湖。該公司希望透過將資料湖中的資料與儲存在 Amazon Aurora MySQL 資料庫中的營運資料結合，在 Amazon QuickSight 中建立視覺化。該公司希望強制執行列級授權，以便公司的行銷團隊只能存取資料庫中列的子集。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

- A. Use Amazon EMR to ingest the data directly from the database to the QuickSight SPICE engine. Include only the required columns.
- B. Use AWS Glue Studio to ingest the data from the database to the S3 data lake. Attach an IAM policy to the QuickSight users to enforce column-level access control. Use Amazon S3 as the data source in QuickSight.

C. Use AWS Glue Elastic Views to create a materialized view for the database in Amazon S3. Create an S3 bucket policy to enforce column-level access control for the QuickSight users. Use Amazon S3 as the data source in QuickSight.

D. Use a Lake Formation blueprint to ingest the data from the database to the S3 data lake. Use Lake Formation to enforce column-level access control for the QuickSight users. Use Amazon Athena as the data source in QuickSight.

【參考答案】D

【題號】question 342

【題目】

A transaction processing company has weekly scripted batch jobs that run on Amazon EC2 instances. The EC2 instances are in an Auto Scaling group. The number of transactions can vary, but the baseline CPU utilization that is noted on each run is at least 60%. The company needs to provision the capacity 30 minutes before the jobs run.

Currently, engineers complete this task by manually modifying the Auto Scaling group parameters. The company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts. The company needs an automated way to modify the Auto Scaling group's desired capacity.

Which solution will meet these requirements with the LEAST operational overhead?

一家事務處理公司每週都有在 Amazon EC2 執行個體上執行的腳本化批次作業。EC2 執行個體位於 Auto Scaling 群組中。交易數量可能會有所不同，但每次運行時記錄的基準 CPU 使用率至少為 60%。公司需在作業開始前 30 分鐘配置產能。

目前，工程師透過手動修改 Auto Scaling 組參數來完成此任務。該公司沒有資源來分析 Auto Scaling 組計數所需的容量趨勢。該公司需要一種自動化的方式來修改 Auto Scaling 組的所需容量。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A. Create a dynamic scaling policy for the Auto Scaling group. Configure the policy to scale based on the CPU utilization metric. Set the target value for the metric to 60%.

B.Create a scheduled scaling policy for the Auto Scaling group. Set the appropriate desired capacity, minimum capacity, and maximum capacity. Set the recurrence to weekly. Set the start time to 30 minutes before the batch jobs run.

C.Create a predictive scaling policy for the Auto Scaling group. Configure the policy to scale based on forecast. Set the scaling metric to CPU utilization. Set the target value for the metric to 60%. In the policy, set the instances to pre-launch 30 minutes before the jobs run.

D.Create an Amazon EventBridge event to invoke an AWS Lambda function when the CPU utilization metric value for the Auto Scaling group reaches 60%. Configure the Lambda function to increase the Auto Scaling group's desired capacity and maximum capacity by 20%.

【參考答案】C

【題號】question 343

【題目】

A solutions architect is designing a company's disaster recovery (DR) architecture. The company has a MySQL database that runs on an Amazon EC2 instance in a private subnet with scheduled backup. The DR design needs to include multiple AWS Regions.

Which solution will meet these requirements with the LEAST operational overhead?

解決方案架構師正在設計公司的災難復原 (DR) 架構。該公司擁有一個 MySQL 資料庫，該資料庫在私有子網路中的 Amazon EC2 執行個體上運行，並具有計劃備份。災難復原設計需要包含多個 AWS 區域。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A.Migrate the MySQL database to multiple EC2 instances. Configure a standby EC2 instance in the DR Region. Turn on replication.

B.Migrate the MySQL database to Amazon RDS. Use a Multi-AZ deployment. Turn on read replication for the primary DB instance in the different Availability Zones.

C.Migrate the MySQL database to an Amazon Aurora global database. Host the primary DB cluster in the primary Region. Host the secondary DB cluster in the DR Region.

D.Store the scheduled backup of the MySQL database in an Amazon S3 bucket that is configured for S3 Cross-Region Replication (CRR). Use the data backup to restore the database in the DR Region.

【參考答案】C

【題號】question 344

【題目】

A company has a Java application that uses Amazon Simple Queue Service (Amazon SQS) to parse messages. The application cannot parse messages that are larger than 256 KB in size. The company wants to implement a solution to give the application the ability to parse messages as large as 50 MB.

Which solution will meet these requirements with the FEWEST changes to the code?

一家公司有一個 Java 應用程式，該應用程式使用 Amazon Simple Queue Service (Amazon SQS) 來解析訊息。應用程式無法解析大小超過 256 KB 的訊息。該公司希望實施一種解決方案，使應用程式能夠解析大至 50 MB 的訊息。

哪種解決方案能夠以最少的程式碼更改滿足這些要求？

【選項】

A.Use the Amazon SQS Extended Client Library for Java to host messages that are larger than 256 KB in Amazon S3.

B.Use Amazon EventBridge to post large messages from the application instead of Amazon SQS.

C.Change the limit in Amazon SQS to handle messages that are larger than 256 KB.

D.Store messages that are larger than 256 KB in Amazon Elastic File System (Amazon EFS). Configure Amazon SQS to reference this location in the messages.

【參考答案】A

【題號】question 345

【題目】

A company wants to restrict access to the content of one of its main web applications and to protect the content by using authorization techniques available on AWS. The company wants to implement a serverless architecture and an authentication solution for fewer than 100 users. The solution needs to integrate with the main web application and serve web content globally. The solution must also scale as the company's user base grows while providing the lowest login latency possible.

Which solution will meet these requirements MOST cost-effectively?

一家公司希望限制對其主要 Web 應用程式之一內容的訪問，並透過使用 AWS 上提供的授權技術來保護內容。該公司希望為少於 100 個用戶實施無伺服器架構和身份驗證解決方案。此解決方案需要與主 Web 應用程式整合並在全球範圍內提供 Web 內容。該解決方案還必須隨著公司用戶群的成長而擴展，同時提供盡可能低的登入延遲。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A. Use Amazon Cognito for authentication. Use Lambda@Edge for authorization. Use Amazon CloudFront to serve the web application globally.

B. Use AWS Directory Service for Microsoft Active Directory for authentication. Use AWS Lambda for authorization. Use an Application Load Balancer to serve the web application globally.

C. Use Amazon Cognito for authentication. Use AWS Lambda for authorization. Use Amazon S3 Transfer Acceleration to serve the web application globally.

D. Use AWS Directory Service for Microsoft Active Directory for authentication. Use Lambda@Edge for authorization. Use AWS Elastic Beanstalk to serve the web application globally.

【參考答案】A

【題號】question 346

【題目】

A company has an aging network-attached storage (NAS) array in its data center. The NAS array presents SMB shares and NFS shares to client workstations. The company does not want to purchase a new NAS array. The company also does not want to incur the cost of renewing the NAS array's support contract. Some of the data is accessed frequently, but much of the data is inactive.

A solutions architect needs to implement a solution that migrates the data to Amazon S3, uses S3 Lifecycle policies, and maintains the same look and feel for the client workstations. The solutions architect has identified AWS Storage Gateway as part of the solution.

Which type of storage gateway should the solutions architect provision to meet these requirements?

一家公司的資料中心內有一個老化的網路附加儲存 (NAS) 陣列。NAS 陣列提供客戶端工作站 SMB 共用和 NFS 共用。該公司不想購買新的 NAS 陣列。該公司也不想承擔續約 NAS 陣列支援合約的費用。某些數據被頻繁訪問，但大部分數據處於非活動狀態。

解決方案架構師需要實施一個解決方案，將資料移轉到 Amazon S3、使用 S3 生命週期策略並保持客戶端工作站的相同外觀。解決方案架構師已將 AWS Storage Gateway 確定為解決方案的一部分。

解決方案架構師應該提供哪種類型的儲存網關來滿足這些要求？

【選項】

A.Volume Gateway

B.Tape Gateway

C.Amazon FSx File Gateway

D.Amazon S3 File Gateway

【參考答案】D

【題號】question 347

【題目】

A company has an application that is running on Amazon EC2 instances. A solutions architect has standardized the company on a particular instance family and various instance sizes based on the current needs of the company.

The company wants to maximize cost savings for the application over the next 3 years. The company needs to be able to change the instance family and sizes in the next 6 months based on application popularity and usage.

Which solution will meet these requirements MOST cost-effectively?

一家公司有一個在 Amazon EC2 執行個體上執行的應用程式。解決方案架構師根據公司目前的需求，在特定實例係列和各種實例大小上對公司進行了標準化。

該公司希望在未來 3 年內最大限度地節省應用成本。該公司需要能夠在未來 6 個月內根據應用程式的受歡迎程度和使用情況更改實例係列和規模。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

- A. Compute Savings Plan
- B. EC2 Instance Savings Plan
- C. Zonal Reserved Instances
- D. Standard Reserved Instances

【參考答案】A

【題號】question 348

【題目】

A company collects data from a large number of participants who use wearable devices. The company stores the data in an Amazon DynamoDB table and uses applications to

analyze the data. The data workload is constant and predictable. The company wants to stay at or below its forecasted budget for DynamoDB.

Which solution will meet these requirements MOST cost-effectively?

一家公司從大量使用穿戴式裝置的參與者收集數據。該公司將資料儲存在 Amazon DynamoDB 表中，並使用應用程式分析資料。資料工作量是恆定且可預測的。該公司希望將 DynamoDB 的預算保持在或低於其預測。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A. Use provisioned mode and DynamoDB Standard-Infrequent Access (DynamoDB Standard-IA). Reserve capacity for the forecasted workload.

B. Use provisioned mode. Specify the read capacity units (RCUs) and write capacity units (WCUs).

C. Use on-demand mode. Set the read capacity units (RCUs) and write capacity units (WCUs) high enough to accommodate changes in the workload.

D. Use on-demand mode. Specify the read capacity units (RCUs) and write capacity units (WCUs) with reserved capacity.

【參考答案】B

【題號】question 349

【題目】

A company stores confidential data in an Amazon Aurora PostgreSQL database in the ap-southeast-3 Region. The database is encrypted with an AWS Key Management Service (AWS KMS) customer managed key. The company was recently acquired and must securely share a backup of the database with the acquiring company's AWS account in ap-southeast-3.

What should a solutions architect do to meet these requirements?

一家公司將機密資料儲存在 ap-southeast-3 區域的 Amazon Aurora PostgreSQL 資料庫中。此資料庫使用 AWS Key Management Service (AWS KMS) 客戶管理的金鑰進行加密。該公司最近被收購，必須與收購公司在 ap-southeast-3 中的 AWS 帳戶安全地共享資料庫備份。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A.Create a database snapshot. Copy the snapshot to a new unencrypted snapshot. Share the new snapshot with the acquiring company's AWS account.

B.Create a database snapshot. Add the acquiring company's AWS account to the KMS key policy. Share the snapshot with the acquiring company's AWS account.

C.Create a database snapshot that uses a different AWS managed KMS key. Add the acquiring company's AWS account to the KMS key alias. Share the snapshot with the acquiring company's AWS account.

D.Create a database snapshot. Download the database snapshot. Upload the database snapshot to an Amazon S3 bucket. Update the S3 bucket policy to allow access from the acquiring company's AWS account.

【參考答案】B

【題號】question 350

【題目】

A company uses a 100 GB Amazon RDS for Microsoft SQL Server Single-AZ DB instance in the us-east-1 Region to store customer transactions. The company needs high availability and automatic recovery for the DB instance.

The company must also run reports on the RDS database several times a year. The report process causes transactions to take longer than usual to post to the customers' accounts. The company needs a solution that will improve the performance of the report process.

Which combination of steps will meet these requirements? (Choose two.)

一家公司在 us-east-1 區域中使用 100 GB Amazon RDS for Microsoft SQL Server 單可用區資料庫執行個體來儲存客戶交易。公司需要資料庫執行個體的高可用性和自動復原。

該公司還必須每年多次在 RDS 資料庫上運行報告。報告流程導致交易需要比平常更長的時間才能過帳到客戶的帳戶。該公司需要一個能夠提高報告流程效能的解決方案。

哪一種步驟組合可以滿足這些要求？（選兩個。）

【選項】

- A.Modify the DB instance from a Single-AZ DB instance to a Multi-AZ deployment.
- B.Take a snapshot of the current DB instance. Restore the snapshot to a new RDS deployment in another Availability Zone.
- C.Create a read replica of the DB instance in a different Availability Zone. Point all requests for reports to the read replica.
- D.Migrate the database to RDS Custom.
- E.Use RDS Proxy to limit reporting requests to the maintenance window.

【參考答案】AC

【題號】question 351

【題目】

A company is moving its data management application to AWS. The company wants to transition to an event-driven architecture. The architecture needs to be more distributed and to use serverless concepts while performing the different aspects of the workflow. The company also wants to minimize operational overhead.

Which solution will meet these requirements?

一家公司正在將其資料管理應用程式遷移到 AWS。該公司希望過渡到事件驅動的架構。此架構需要更加分散式，並在執行工作流程的不同方面時使用無伺服器概念。該公司還希望最大限度地減少營運費用。

哪種解決方案可以滿足這些要求？

【選項】

- A. Build out the workflow in AWS Glue. Use AWS Glue to invoke AWS Lambda functions to process the workflow steps.
- B. Build out the workflow in AWS Step Functions. Deploy the application on Amazon EC2 instances. Use Step Functions to invoke the workflow steps on the EC2 instances.
- C. Build out the workflow in Amazon EventBridge. Use EventBridge to invoke AWS Lambda functions on a schedule to process the workflow steps.
- D. Build out the workflow in AWS Step Functions. Use Step Functions to create a state machine. Use the state machine to invoke AWS Lambda functions to process the workflow steps.

【參考答案】D

【題號】question 352

【題目】

A company is designing the network for an online multi-player game. The game uses the UDP networking protocol and will be deployed in eight AWS Regions. The network architecture needs to minimize latency and packet loss to give end users a high-quality gaming experience.

Which solution will meet these requirements?

一家公司正在為線上多人遊戲設計網路。遊戲使用 UDP 網路協議，將部署在八個 AWS 區域。網路架構需要最大限度地減少延遲和丟包，為最終用戶提供高品質的遊戲體驗。

哪種解決方案可以滿足這些要求？

【選項】

- A. Setup a transit gateway in each Region. Create inter-Region peering attachments between each transit gateway.
- B. Set up AWS Global Accelerator with UDP listeners and endpoint groups in each Region.
- C. Set up Amazon CloudFront with UDP turned on. Configure an origin in each Region.
- D. Set up a VPC peering mesh between each Region. Turn on UDP for each VPC.

【參考答案】B

【題號】question 353

【題目】

A company hosts a three-tier web application on Amazon EC2 instances in a single Availability Zone. The web application uses a self-managed MySQL database that is hosted on an EC2 instance to store data in an Amazon Elastic Block Store (Amazon EBS) volume. The MySQL database currently uses a 1 TB Provisioned IOPS SSD (io2) EBS volume. The company expects traffic of 1,000 IOPS for both reads and writes at peak traffic.

The company wants to minimize any disruptions, stabilize performance, and reduce costs while retaining the capacity for double the IOPS. The company wants to move the database tier to a fully managed solution that is highly available and fault tolerant.

Which solution will meet these requirements MOST cost-effectively?

一家公司在單一可用區的 Amazon EC2 執行個體上託管一個三層 Web 應用程式。此 Web 應用程式使用託管在 EC2 執行個體上的自我管理 MySQL 資料庫將資料儲存在 Amazon Elastic Block Store (Amazon EBS) 磁碟區中。MySQL 資料庫目前使用 1 TB 預先設定 IOPS SSD (io2) EBS 磁碟區。該公司預計峰值流量時的讀取和寫入流量均為 1,000 IOPS。

該公司希望最大限度地減少任何中斷、穩定性能並降低成本，同時保留雙倍 IOPS 的容量。該公司希望將資料庫層遷移到高度可用且容錯的完全託管解決方案。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

- A. Use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with an io2 Block Express EBS volume.
- B. Use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with a General Purpose SSD (gp2) EBS volume.
- C. Use Amazon S3 Intelligent-Tiering access tiers.
- D. Use two large EC2 instances to host the database in active-passive mode.

【參考答案】B

【題號】question 354

【題目】

A company hosts a serverless application on AWS. The application uses Amazon API Gateway, AWS Lambda, and an Amazon RDS for PostgreSQL database. The company notices an increase in application errors that result from database connection timeouts during times of peak traffic or unpredictable traffic. The company needs a solution that reduces the application failures with the least amount of change to the code.

What should a solutions architect do to meet these requirements?

一家公司在 AWS 上託管無伺服器應用程式。該應用程式使用 Amazon API Gateway、AWS Lambda 和 Amazon RDS for PostgreSQL 資料庫。該公司注意到，在高峰流量或不可預測的流量期間，由於資料庫連接逾時而導致的應用程式錯誤增加。該公司需要一種解決方案，能夠以最少的程式碼更改來減少應用程式故障。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

- A.Reduce the Lambda concurrency rate.
- B.Enable RDS Proxy on the RDS DB instance.
- C.Resize the RDS DB instance class to accept more connections.
- D.Migrate the database to Amazon DynamoDB with on-demand scaling.

【參考答案】B

【題號】question 355

【題目】

A company is migrating an old application to AWS. The application runs a batch job every hour and is CPU intensive. The batch job takes 15 minutes on average with an on-premises server. The server has 64 virtual CPU (vCPU) and 512 GiB of memory.

Which solution will run the batch job within 15 minutes with the LEAST operational overhead?

一家公司正在將舊應用程式遷移到 AWS。該應用程式每小時運行一個批次作業，並且是 CPU 密集型的。使用本機伺服器時，批次作業平均需要 15 分鐘。該伺服器具有 64 個虛擬 CPU (vCPU) 和 512 GiB 記憶體。

哪種解決方案能夠在 15 分鐘內以最少的營運開銷執行批次作業？

【選項】

- A. Use AWS Lambda with functional scaling.
- B. Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate.
- C. Use Amazon Lightsail with AWS Auto Scaling.
- D. Use AWS Batch on Amazon EC2.

【參考答案】D

【題號】question 356

【題目】

A company stores its data objects in Amazon S3 Standard storage. A solutions architect has found that 75% of the data is rarely accessed after 30 days. The company needs all the data to remain immediately accessible with the same high availability and resiliency, but the company wants to minimize storage costs.

Which storage solution will meet these requirements?

一家公司將其資料物件儲存在 Amazon S3 標準儲存中。解決方案架構師發現 75% 的資料在 30 天後很少被存取。該公司需要所有數據都能立即訪問，並具有相同的高可用性和彈性，但公司希望最大限度地降低儲存成本。

哪種儲存解決方案可以滿足這些要求？

【選項】

- A.Move the data objects to S3 Glacier Deep Archive after 30 days.
- B.Move the data objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days.
- C.Move the data objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 30 days.
- D.Move the data objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) immediately.

【參考答案】B

【題號】question 357

【題目】

A gaming company is moving its public scoreboard from a data center to the AWS Cloud. The company uses Amazon EC2 Windows Server instances behind an Application Load Balancer to host its dynamic application. The company needs a highly available storage solution for the application. The application consists of static files and dynamic server-side code.

Which combination of steps should a solutions architect take to meet these requirements? (Choose two.)

一家遊戲公司正在將其公共記分牌從資料中心遷移到 AWS 雲端。該公司在應用程式負載平衡器後面使用 Amazon EC2 Windows Server 執行個體來託管其動態應用程式。該公司需要為應用程式提供高度可用的儲存解決方案。該應用程式由靜態檔案和動態伺服器端程式碼組成。

解決方案架構師應該採取哪些步驟組合來滿足這些要求？（選兩個。）

【選項】

- A.Store the static files on Amazon S3. Use Amazon CloudFront to cache objects at the edge.

B.Store the static files on Amazon S3. Use Amazon ElastiCache to cache objects at the edge.

C.Store the server-side code on Amazon Elastic File System (Amazon EFS). Mount the EFS volume on each EC2 instance to share the files.

D.Store the server-side code on Amazon FSx for Windows File Server. Mount the FSx for Windows File Server volume on each EC2 instance to share the files.

E.Store the server-side code on a General Purpose SSD (gp2) Amazon Elastic Block Store (Amazon EBS) volume. Mount the EBS volume on each EC2 instance to share the files.

【參考答案】AD

【題號】question 358

【題目】

A social media company runs its application on Amazon EC2 instances behind an Application Load Balancer (ALB). The ALB is the origin for an Amazon CloudFront distribution. The application has more than a billion images stored in an Amazon S3 bucket and processes thousands of images each second. The company wants to resize the images dynamically and serve appropriate formats to clients.

Which solution will meet these requirements with the LEAST operational overhead?

一家社群媒體公司在應用程式負載平衡器 (ALB) 後面的 Amazon EC2 執行個體上執行其應用程式。ALB 是 Amazon CloudFront 發行版的起源。該應用程式在 Amazon S3 儲存桶中儲存了超過十億張影像，每秒處理數千張影像。該公司希望動態調整圖像大小並為客戶提供適當的格式。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A.Install an external image management library on an EC2 instance. Use the image management library to process the images.

B.Create a CloudFront origin request policy. Use the policy to automatically resize images and to serve the appropriate format based on the User-Agent HTTP header in the request.

C.Use a Lambda@Edge function with an external image management library. Associate the Lambda@Edge function with the CloudFront behaviors that serve the images.

D.Create a CloudFront response headers policy. Use the policy to automatically resize images and to serve the appropriate format based on the User-Agent HTTP header in the request.

【參考答案】C

【題號】question 359

【題目】

A hospital needs to store patient records in an Amazon S3 bucket. The hospital's compliance team must ensure that all protected health information (PHI) is encrypted in transit and at rest. The compliance team must administer the encryption key for data at rest.

Which solution will meet these requirements?

醫院需要將病患記錄儲存在 Amazon S3 儲存桶中。醫院的合規團隊必須確保所有受保護的健康資訊 (PHI) 在傳輸和靜態時都已加密。合規團隊必須管理靜態資料的加密金鑰。

哪種解決方案可以滿足這些要求？

【選項】

A.Create a public SSL/TLS certificate in AWS Certificate Manager (ACM). Associate the certificate with Amazon S3. Configure default encryption for each S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Assign the compliance team to manage the KMS keys.

B.Use the aws:SecureTransport condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Configure default encryption for each S3 bucket to use server-side encryption with S3 managed encryption keys (SSE-S3). Assign the compliance team to manage the SSE-S3 keys.

C. Use the aws:SecureTransport condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Configure default encryption for each S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Assign the compliance team to manage the KMS keys.

D. Use the aws:SecureTransport condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Use Amazon Macie to protect the sensitive data that is stored in Amazon S3. Assign the compliance team to manage Macie.

【參考答案】C

【題號】question 360

【題目】

A company uses Amazon API Gateway to run a private gateway with two REST APIs in the same VPC. The BuyStock RESTful web service calls the CheckFunds RESTful web service to ensure that enough funds are available before a stock can be purchased. The company has noticed in the VPC flow logs that the BuyStock RESTful web service calls the CheckFunds RESTful web service over the internet instead of through the VPC. A solutions architect must implement a solution so that the APIs communicate through the VPC.

Which solution will meet these requirements with the FEWEST changes to the code?

一家公司使用 Amazon API Gateway 在同一 VPC 中執行具有兩個 REST API 的私人網關。BuyStock RESTful Web 服務呼叫 CheckFunds RESTful Web 服務，以確保在購買股票之前有足夠的可用資金。該公司在 VPC 流程日誌中註意到，BuyStock RESTful Web 服務透過互聯網而不是透過 VPC 呼叫 CheckFunds RESTful Web 服務。解決方案架構師必須實作解決方案，以便 API 透過 VPC 進行通訊。

哪種解決方案能夠以最少的程式碼更改滿足這些要求？

【選項】

A. Add an X-API-Key header in the HTTP header for authorization.

B. Use an interface endpoint.

C. Use a gateway endpoint.

D.Add an Amazon Simple Queue Service (Amazon SQS) queue between the two REST APIs.

【參考答案】B

【題號】question 361

【題目】

A company hosts a multiplayer gaming application on AWS. The company wants the application to read data with sub-millisecond latency and run one-time queries on historical data.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司在 AWS 上託管多人遊戲應用程式。該公司希望應用程式能夠以亞毫秒延遲讀取資料並對歷史資料運行一次性查詢。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A.Use Amazon RDS for data that is frequently accessed. Run a periodic custom script to export the data to an Amazon S3 bucket.

B.Store the data directly in an Amazon S3 bucket. Implement an S3 Lifecycle policy to move older data to S3 Glacier Deep Archive for long-term storage. Run one-time queries on the data in Amazon S3 by using Amazon Athena.

C.Use Amazon DynamoDB with DynamoDB Accelerator (DAX) for data that is frequently accessed. Export the data to an Amazon S3 bucket by using DynamoDB table export. Run one-time queries on the data in Amazon S3 by using Amazon Athena.

D.Use Amazon DynamoDB for data that is frequently accessed. Turn on streaming to Amazon Kinesis Data Streams. Use Amazon Kinesis Data Firehose to read the data from Kinesis Data Streams. Store the records in an Amazon S3 bucket.

【參考答案】C

【題號】question 362

【題目】

A company uses a payment processing system that requires messages for a particular payment ID to be received in the same order that they were sent. Otherwise, the payments might be processed incorrectly.

Which actions should a solutions architect take to meet this requirement? (Choose two.)

本公司使用的支付處理系統要求依照傳送訊息的順序接收特定支付 ID 的訊息。否則，付款可能會被錯誤處理。

解決方案架構師應該採取哪些行動來滿足此要求？（選兩個。）

【選項】

A. Write the messages to an Amazon DynamoDB table with the payment ID as the partition key.

B. Write the messages to an Amazon Kinesis data stream with the payment ID as the partition key.

C. Write the messages to an Amazon ElastiCache for Memcached cluster with the payment ID as the key.

D. Write the messages to an Amazon Simple Queue Service (Amazon SQS) queue. Set the message attribute to use the payment ID.

E. Write the messages to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Set the message group to use the payment ID.

【參考答案】BE

【題號】question 363

【題目】

A company is building a game system that needs to send unique events to separate leaderboard, matchmaking, and authentication services concurrently. The company needs an AWS event-driven system that guarantees the order of the events.

Which solution will meet these requirements?

一家公司正在建立一個遊戲系統，需要同時將獨特的事件發送到單獨的排行榜、配對和身份驗證服務。該公司需要一個 AWS 事件驅動系統來保證事件的順序。

哪種解決方案可以滿足這些要求？

【選項】

A.Amazon EventBridge event bus

B.Amazon Simple Notification Service (Amazon SNS) FIFO topics

C.Amazon Simple Notification Service (Amazon SNS) standard topics

D.Amazon Simple Queue Service (Amazon SQS) FIFO queues

【參考答案】B

【題號】question 364

【題目】

A hospital is designing a new application that gathers symptoms from patients. The hospital has decided to use Amazon Simple Queue Service (Amazon SQS) and Amazon Simple Notification Service (Amazon SNS) in the architecture.

A solutions architect is reviewing the infrastructure design. Data must be encrypted at rest and in transit. Only authorized personnel of the hospital should be able to access the data.

Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

一家醫院正在設計一種新的應用程序，用於收集患者的症狀。該醫院決定在架構中使用 Amazon Simple Queue Service (Amazon SQS) 和 Amazon Simple notification Service (Amazon SNS)。

解決方案架構師正在審查基礎設施設計。資料在靜態和傳輸過程中必須進行加密。只有醫院的授權人員才能存取資料。

解決方案架構師應該採取哪些步驟組合來滿足這些要求？（選兩個。）

【選項】

A.Turn on server-side encryption on the SQS components. Update the default key policy to restrict key usage to a set of authorized principals.

B.Turn on server-side encryption on the SNS components by using an AWS Key Management Service (AWS KMS) customer managed key. Apply a key policy to restrict key usage to a set of authorized principals.

C.Turn on encryption on the SNS components. Update the default key policy to restrict key usage to a set of authorized principals. Set a condition in the topic policy to allow only encrypted connections over TLS.

D.Turn on server-side encryption on the SQS components by using an AWS Key Management Service (AWS KMS) customer managed key. Apply a key policy to restrict key usage to a set of authorized principals. Set a condition in the queue policy to allow only encrypted connections over TLS.

E.Turn on server-side encryption on the SQS components by using an AWS Key Management Service (AWS KMS) customer managed key. Apply an IAM policy to restrict key usage to a set of authorized principals. Set a condition in the queue policy to allow only encrypted connections over TLS.

【參考答案】BD

【題號】question 365

【題目】

A company runs a web application that is backed by Amazon RDS. A new database administrator caused data loss by accidentally editing information in a database table. To help recover from this type of incident, the company wants the ability to restore the database to its state from 5 minutes before any change within the last 30 days.

Which feature should the solutions architect include in the design to meet this requirement?

一家公司運行由 Amazon RDS 支援的 Web 應用程式。一位新的資料庫管理員因意外編輯資料庫表中的資訊而導致資料遺失。為了幫助從此類事件中恢復，該公司希望能夠將資料庫恢復到過去 30 天內發生任何更改之前 5 分鐘的狀態。

解決方案架構師應該在設計中包含哪些功能來滿足此要求？

【選項】

- A.Read replicas
- B.Manual snapshots
- C.Automated backups
- D.Multi-AZ deployments

【參考答案】C

【題號】question 366

【題目】

A company's web application consists of an Amazon API Gateway API in front of an AWS Lambda function and an Amazon DynamoDB database. The Lambda function handles the business logic, and the DynamoDB table hosts the data. The application uses Amazon Cognito user pools to identify the individual users of the application. A solutions architect needs to update the application so that only users who have a subscription can access premium content.

Which solution will meet this requirement with the LEAST operational overhead?

該公司的 Web 應用程式由 AWS Lambda 函數前面的 Amazon API Gateway API 和 Amazon DynamoDB 資料庫組成。Lambda 函數處理業務邏輯，DynamoDB 資料庫託管資料。該應用程式使用 Amazon Cognito 用戶池來識別應用程式的各個用戶。解決方案架構師需要更新應用程式，以便只有訂閱的用戶才能存取優質內容。

哪種解決方案能夠以最少的營運開銷滿足這項要求？

【選項】

- A.Enable API caching and throttling on the API Gateway API.
- B.Set up AWS WAF on the API Gateway API. Create a rule to filter users who have a subscription.
- C.Apply fine-grained IAM permissions to the premium content in the DynamoDB table.
- D.Implement API usage plans and API keys to limit the access of users who do not have a subscription.

【參考答案】D

【題號】question 367

【題目】

A company is using Amazon Route 53 latency-based routing to route requests to its UDP-based application for users around the world. The application is hosted on redundant servers in the company's on-premises data centers in the United States, Asia, and Europe. The company's compliance requirements state that the application must be hosted on premises. The company wants to improve the performance and availability of the application.

What should a solutions architect do to meet these requirements?

一家公司正在使用 Amazon Route 53 基於延遲的路由將世界各地用戶的請求路由到其基於 UDP 的應用程式。該應用程式託管在該公司位於美國、亞洲和歐洲的本地資料中心的冗餘伺服器上。該公司的合規性要求規定應用程式必須在本地託管。該公司希望提高應用程式的效能和可用性。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

- A.Configure three Network Load Balancers (NLBs) in the three AWS Regions to address the on-premises endpoints. Create an accelerator by using AWS Global Accelerator, and

register the NLBs as its endpoints. Provide access to the application by using a CNAME that points to the accelerator DNS.

B.Configure three Application Load Balancers (ALBs) in the three AWS Regions to address the on-premises endpoints. Create an accelerator by using AWS Global Accelerator, and register the ALBs as its endpoints. Provide access to the application by using a CNAME that points to the accelerator DNS.

C.Configure three Network Load Balancers (NLBs) in the three AWS Regions to address the on-premises endpoints. In Route 53, create a latency-based record that points to the three NLBs, and use it as an origin for an Amazon CloudFront distribution. Provide access to the application by using a CNAME that points to the CloudFront DNS.

D.Configure three Application Load Balancers (ALBs) in the three AWS Regions to address the on-premises endpoints. In Route 53, create a latency-based record that points to the three ALBs, and use it as an origin for an Amazon CloudFront distribution. Provide access to the application by using a CNAME that points to the CloudFront DNS.

【參考答案】A

【題號】question 368

【題目】

A solutions architect wants all new users to have specific complexity requirements and mandatory rotation periods for IAM user passwords.

What should the solutions architect do to accomplish this?

解決方案架構師希望所有新使用者都有特定的複雜性要求和 IAM 使用者密碼的強制輪替期。

解決方案架構師應該做什麼來實現這個目標？

【選項】

A.Set an overall password policy for the entire AWS account.

B.Set a password policy for each IAM user in the AWS account.

C.Use third-party vendor software to set password requirements.

D.Attach an Amazon CloudWatch rule to the Create_newuser event to set the password with the appropriate requirements.

【參考答案】A

【題號】question 369

【題目】

A company has migrated an application to Amazon EC2 Linux instances. One of these EC2 instances runs several 1-hour tasks on a schedule. These tasks were written by different teams and have no common programming language. The company is concerned about performance and scalability while these tasks run on a single instance. A solutions architect needs to implement a solution to resolve these concerns.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司已將應用程式遷移到 Amazon EC2 Linux 執行個體。其中一個 EC2 執行個體會依照計畫執行多個 1 小時的任務。這些任務是由不同的團隊編寫的，沒有通用的程式語言。該公司擔心這些任務在單一實例上運行時的效能和可擴展性。解決方案架構師需要實作一個解決方案來解決這些問題。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A.Use AWS Batch to run the tasks as jobs. Schedule the jobs by using Amazon EventBridge (Amazon CloudWatch Events).

B.Convert the EC2 instance to a container. Use AWS App Runner to create the container on demand to run the tasks as jobs.

C.Copy the tasks into AWS Lambda functions. Schedule the Lambda functions by using Amazon EventBridge (Amazon CloudWatch Events).

D.Create an Amazon Machine Image (AMI) of the EC2 instance that runs the tasks. Create an Auto Scaling group with the AMI to run multiple copies of the instance.

【參考答案】A

【題號】question 370

【題目】

A company runs a public three-tier web application in a VPC. The application runs on Amazon EC2 instances across multiple Availability Zones. The EC2 instances that run in private subnets need to communicate with a license server over the internet. The company needs a managed solution that minimizes operational maintenance.

Which solution meets these requirements?

一家公司在 VPC 中運行公共三層 Web 應用程式。該應用程式在跨多個可用區的 Amazon EC2 執行個體上執行。在私有子網路中執行的 EC2 執行個體需要透過 Internet 與許可證伺服器進行通訊。該公司需要一個可最大限度減少營運維護的託管解決方案。

哪種解決方案滿足這些要求？

【選項】

A.Provision a NAT instance in a public subnet. Modify each private subnet's route table with a default route that points to the NAT instance.

B.Provision a NAT instance in a private subnet. Modify each private subnet's route table with a default route that points to the NAT instance.

C.Provision a NAT gateway in a public subnet. Modify each private subnet's route table with a default route that points to the NAT gateway.

D.Provision a NAT gateway in a private subnet. Modify each private subnet's route table with a default route that points to the NAT gateway.

【參考答案】C

【題號】question 371

【題目】

A company needs to create an Amazon Elastic Kubernetes Service (Amazon EKS) cluster to host a digital media streaming application. The EKS cluster will use a managed node group that is backed by Amazon Elastic Block Store (Amazon EBS) volumes for storage. The company must encrypt all data at rest by using a customer managed key that is stored in AWS Key Management Service (AWS KMS).

Which combination of actions will meet this requirement with the LEAST operational overhead? (Choose two.)

一家公司需要建立 Amazon Elastic Kubernetes Service (Amazon EKS) 叢集來託管數位媒體串流應用程式。EKS 叢集將使用由 Amazon Elastic Block Store (Amazon EBS) 磁碟區支援的託管節點群組進行儲存。公司必須使用儲存在 AWS Key Management Service (AWS KMS) 中的客戶託管金鑰對所有靜態資料進行加密。

哪種操作組合能夠以最少的營運開銷滿足此要求？（選兩個。）

【選項】

- A. Use a Kubernetes plugin that uses the customer managed key to perform data encryption.
- B. After creation of the EKS cluster, locate the EBS volumes. Enable encryption by using the customer managed key.
- C. Enable EBS encryption by default in the AWS Region where the EKS cluster will be created. Select the customer managed key as the default key.
- D. Create the EKS cluster. Create an IAM role that has a policy that grants permission to the customer managed key. Associate the role with the EKS cluster.
- E. Store the customer managed key as a Kubernetes secret in the EKS cluster. Use the customer managed key to encrypt the EBS volumes.

【參考答案】CD

【題號】question 372

【題目】

A company wants to migrate an Oracle database to AWS. The database consists of a single table that contains millions of geographic information systems (GIS) images that are high resolution and are identified by a geographic code.

When a natural disaster occurs, tens of thousands of images get updated every few minutes. Each geographic code has a single image or row that is associated with it. The company wants a solution that is highly available and scalable during such events.

Which solution meets these requirements MOST cost-effectively?

一家公司想要將 Oracle 資料庫遷移到 AWS。該資料庫由一個表組成，其中包含數百萬個高解析度的地理資訊系統 (GIS) 影像，並透過地理代碼進行識別。

當自然災害發生時，每隔幾分鐘就會更新數萬張影像。每個地理代碼都有一個與其關聯的圖像或行。該公司希望有一個在此類事件期間具有高可用性和可擴展性的解決方案。

哪種解決方案最經濟有效率地滿足這些要求？

【選項】

A.Store the images and geographic codes in a database table. Use Oracle running on an Amazon RDS Multi-AZ DB instance.

B.Store the images in Amazon S3 buckets. Use Amazon DynamoDB with the geographic code as the key and the image S3 URL as the value.

C.Store the images and geographic codes in an Amazon DynamoDB table. Configure DynamoDB Accelerator (DAX) during times of high load.

D.Store the images in Amazon S3 buckets. Store geographic codes and image S3 URLs in a database table. Use Oracle running on an Amazon RDS Multi-AZ DB instance.

【參考答案】B

【題號】question 373

【題目】

A company has an application that collects data from IoT sensors on automobiles. The data is streamed and stored in Amazon S3 through Amazon Kinesis Data Firehose. The

data produces trillions of S3 objects each year. Each morning, the company uses the data from the previous 30 days to retrain a suite of machine learning (ML) models.

Four times each year, the company uses the data from the previous 12 months to perform analysis and train other ML models. The data must be available with minimal delay for up to 1 year. After 1 year, the data must be retained for archival purposes.

Which storage solution meets these requirements MOST cost-effectively?

一家公司有一個應用程式，可以從汽車上的物聯網感測器收集數據。資料透過 Amazon Kinesis Data Firehose 串流並儲存在 Amazon S3 中。這些數據每年產生數萬億個 S3 物件。每天早上，該公司都會使用過去 30 天的資料重新訓練一套機器學習 (ML) 模型。

該公司每年四次使用前 12 個月的數據進行分析並訓練其他機器學習模型。資料的可用時間必須最短，延遲時間最長為 1 年。一年後，資料必須保留以供存檔。

哪種儲存解決方案最經濟有效率地滿足這些要求？

【選項】

A. Use the S3 Intelligent-Tiering storage class. Create an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 1 year.

B. Use the S3 Intelligent-Tiering storage class. Configure S3 Intelligent-Tiering to automatically move objects to S3 Glacier Deep Archive after 1 year.

C. Use the S3 Standard-Infrequent Access (S3 Standard-IA) storage class. Create an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 1 year.

D. Use the S3 Standard storage class. Create an S3 Lifecycle policy to transition objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days, and then to S3 Glacier Deep Archive after 1 year.

【參考答案】D

【題號】question 374

【題目】

A company is running several business applications in three separate VPCs within the us-east-1 Region. The applications must be able to communicate between VPCs. The

applications also must be able to consistently send hundreds of gigabytes of data each day to a latency-sensitive application that runs in a single on-premises data center.

A solutions architect needs to design a network connectivity solution that maximizes cost-effectiveness.

Which solution meets these requirements?

一家公司正在 us-east-1 區域內的三個獨立的 VPC 中運行多個業務應用程式。應用程式必須能夠在 VPC 之間進行通訊。應用程式還必須能夠每天持續將數百 GB 的資料傳送到在單一本
地資料中心運行的延遲敏感型應用程式。

解決方案架構師需要設計一個能夠最大限度提高成本效益的網路連接解決方案。

哪種解決方案滿足這些要求？

【選項】

A. Configure three AWS Site-to-Site VPN connections from the data center to AWS. Establish connectivity by configuring one VPN connection for each VPC.

B. Launch a third-party virtual network appliance in each VPC. Establish an IPsec VPN tunnel between the data center and each virtual appliance.

C. Set up three AWS Direct Connect connections from the data center to a Direct Connect gateway in us-east-1. Establish connectivity by configuring each VPC to use one of the Direct Connect connections.

D. Set up one AWS Direct Connect connection from the data center to AWS. Create a transit gateway, and attach each VPC to the transit gateway. Establish connectivity between the Direct Connect connection and the transit gateway.

【參考答案】D

【題號】question 375

【題目】

An ecommerce company is building a distributed application that involves several serverless functions and AWS services to complete order-processing tasks. These tasks require manual approvals as part of the workflow. A solutions architect needs to design

an architecture for the order-processing application. The solution must be able to combine multiple AWS Lambda functions into responsive serverless applications. The solution also must orchestrate data and services that run on Amazon EC2 instances, containers, or on-premises servers.

Which solution will meet these requirements with the LEAST operational overhead?

一家電子商務公司正在建立一個分散式應用程式，其中涉及多個無伺服器功能和 AWS 服務來完成訂單處理任務。作為工作流程的一部分，這些任務需要手動批准。解決方案架構師需要為訂單處理應用程式設計一個架構。此解決方案必須能夠將多個 AWS Lambda 函數組合到響應式無伺服器應用程式中。此解決方案還必須編排在 Amazon EC2 執行個體、容器或本機伺服器上執行的資料和服務。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

- A. Use AWS Step Functions to build the application.
- B. Integrate all the application components in an AWS Glue job.
- C. Use Amazon Simple Queue Service (Amazon SQS) to build the application.
- D. Use AWS Lambda functions and Amazon EventBridge events to build the application.

【參考答案】A

【題號】question 376

【題目】

A company has launched an Amazon RDS for MySQL DB instance. Most of the connections to the database come from serverless applications. Application traffic to the database changes significantly at random intervals. At times of high demand, users report that their applications experience database connection rejection errors.

Which solution will resolve this issue with the LEAST operational overhead?

一家公司推出了 Amazon RDS for MySQL 資料庫執行個體。大多數資料庫連接來自無伺服器應用程式。資料庫的應用程式流量以隨機間隔顯著變化。在需求量很大時，用戶報告他們的應用程式遇到資料庫連線拒絕錯誤。

哪種解決方案能夠以最少的營運開銷解決此問題？

【選項】

A.Create a proxy in RDS Proxy. Configure the users' applications to use the DB instance through RDS Proxy.

B.Deploy Amazon ElastiCache for Memcached between the users' applications and the DB instance.

C.Migrate the DB instance to a different instance class that has higher I/O capacity. Configure the users' applications to use the new DB instance.

D.Configure Multi-AZ for the DB instance. Configure the users' applications to switch between the DB instances.

【參考答案】A

【題號】question 377

【題目】

A company recently deployed a new auditing system to centralize information about operating system versions, patching, and installed software for Amazon EC2 instances. A solutions architect must ensure all instances provisioned through EC2 Auto Scaling groups successfully send reports to the auditing system as soon as they are launched and terminated.

Which solution achieves these goals MOST efficiently?

一家公司最近部署了一個新的審核系統，以集中有關 Amazon EC2 執行個體的作業系統版本、修補和安裝軟體的資訊。解決方案架構師必須確保透過 EC2 Auto Scaling 群組配置的所有執行個體在啟動和終止後立即成功將報告傳送至審核系統。

哪一種解決方案最有效地實現這些目標？

【選項】

A. Use a scheduled AWS Lambda function and run a script remotely on all EC2 instances to send data to the audit system.

B. Use EC2 Auto Scaling lifecycle hooks to run a custom script to send data to the audit system when instances are launched and terminated.

C. Use an EC2 Auto Scaling launch configuration to run a custom script through user data to send data to the audit system when instances are launched and terminated.

D. Run a custom script on the instance operating system to send data to the audit system. Configure the script to be invoked by the EC2 Auto Scaling group when the instance starts and is terminated.

【參考答案】B

【題號】question 378

【題目】

A company is developing a real-time multiplayer game that uses UDP for communications between the client and servers in an Auto Scaling group. Spikes in demand are anticipated during the day, so the game server platform must adapt accordingly. Developers want to store gamer scores and other non-relational data in a database solution that will scale without intervention.

Which solution should a solutions architect recommend?

一家公司正在開發一款即時多人遊戲，該遊戲使用 UDP 在 Auto Scaling 群組中的客戶端和伺服器之間進行通訊。預計白天的需求會激增，因此遊戲伺服器平台必須做出相應調整。開發人員希望將玩家分數和其他非關聯式資料儲存在無需幹預即可擴展的資料庫解決方案中。

解決方案架構師應該推薦哪一種解決方案？

【選項】

A. Use Amazon Route 53 for traffic distribution and Amazon Aurora Serverless for data storage.

B. Use a Network Load Balancer for traffic distribution and Amazon DynamoDB on-demand for data storage.

C. Use a Network Load Balancer for traffic distribution and Amazon Aurora Global Database for data storage.

D. Use an Application Load Balancer for traffic distribution and Amazon DynamoDB global tables for data storage.

【參考答案】B

【題號】question 379

【題目】

A company hosts a frontend application that uses an Amazon API Gateway API backend that is integrated with AWS Lambda. When the API receives requests, the Lambda function loads many libraries. Then the Lambda function connects to an Amazon RDS database, processes the data, and returns the data to the frontend application. The company wants to ensure that response latency is as low as possible for all its users with the fewest number of changes to the company's operations.

Which solution will meet these requirements?

一家公司託管一個使用與 AWS Lambda 整合的 Amazon API Gateway API 後端的前端應用程式。當 API 收到請求時，Lambda 函數會載入許多函式庫。然後，Lambda 函數連接到 Amazon RDS 資料庫，處理數據，並將數據傳回前端應用程式。該公司希望確保所有用戶的回應延遲盡可能低，同時對公司營運的變更最少。

哪種解決方案可以滿足這些要求？

【選項】

A. Establish a connection between the frontend application and the database to make queries faster by bypassing the API.

B. Configure provisioned concurrency for the Lambda function that handles the requests.

C. Cache the results of the queries in Amazon S3 for faster retrieval of similar datasets.

D. Increase the size of the database to increase the number of connections Lambda can establish at one time.

【參考答案】B

【題號】question 380

【題目】

A company is migrating its on-premises workload to the AWS Cloud. The company already uses several Amazon EC2 instances and Amazon RDS DB instances. The company wants a solution that automatically starts and stops the EC2 instances and DB instances outside of business hours. The solution must minimize cost and infrastructure maintenance.

Which solution will meet these requirements?

一家公司正在將其本地工作負載遷移到 AWS 雲端。該公司已使用多個 Amazon EC2 執行個體和 Amazon RDS 資料庫執行個體。該公司需要一個能夠在工作時間之外自動啟動和停止 EC2 執行個體和資料庫執行個體的解決方案。此解決方案必須最大限度地減少成本和基礎設施維護。

哪種解決方案可以滿足這些要求？

【選項】

A.Scale the EC2 instances by using elastic resize. Scale the DB instances to zero outside of business hours.

B.Explore AWS Marketplace for partner solutions that will automatically start and stop the EC2 instances and DB instances on a schedule.

C.Launch another EC2 instance. Configure a crontab schedule to run shell scripts that will start and stop the existing EC2 instances and DB instances on a schedule.

D.Create an AWS Lambda function that will start and stop the EC2 instances and DB instances. Configure Amazon EventBridge to invoke the Lambda function on a schedule.

【參考答案】D

【題號】question 381

【題目】

A company hosts a three-tier web application that includes a PostgreSQL database. The database stores the metadata from documents. The company searches the metadata for key terms to retrieve documents that the company reviews in a report each month. The documents are stored in Amazon S3. The documents are usually written only once, but they are updated frequently.

The reporting process takes a few hours with the use of relational queries. The reporting process must not prevent any document modifications or the addition of new documents. A solutions architect needs to implement a solution to speed up the reporting process.

Which solution will meet these requirements with the LEAST amount of change to the application code?

一家公司託管一個包含 PostgreSQL 資料庫的三層 Web 應用程式。資料庫儲存文檔的元資料。該公司在元資料中搜尋關鍵術語，以檢索該公司每月在報告中審查的文件。這些文件儲存在 Amazon S3 中。這些文件通常只編寫一次，但會經常更新。

使用關係查詢，報告過程需要幾個小時。報告過程不得阻止任何文件修改或新增文件。解決方案架構師需要實施解決方案來加快報告流程。

哪種解決方案能夠以最少的應用程式程式碼變更來滿足這些要求？

【選項】

A.Set up a new Amazon DocumentDB (with MongoDB compatibility) cluster that includes a read replica. Scale the read replica to generate the reports.

B.Set up a new Amazon Aurora PostgreSQL DB cluster that includes an Aurora Replica. Issue queries to the Aurora Replica to generate the reports.

C.Set up a new Amazon RDS for PostgreSQL Multi-AZ DB instance. Configure the reporting module to query the secondary RDS node so that the reporting module does not affect the primary node.

D.Set up a new Amazon DynamoDB table to store the documents. Use a fixed write capacity to support new document entries. Automatically scale the read capacity to support the reports.

【參考答案】B

【題號】question 382

【題目】

A company has a three-tier application on AWS that ingests sensor data from its users' devices. The traffic flows through a Network Load Balancer (NLB), then to Amazon EC2 instances for the web tier, and finally to EC2 instances for the application tier. The application tier makes calls to a database.

What should a solutions architect do to improve the security of the data in transit?

一家公司在 AWS 上有一個三層應用程序，可以從用戶的設備中獲取感測器數據。流量流經網路負載平衡器 (NLB)，然後流向 Web 層的 Amazon EC2 實例，最後流向應用程式層的 EC2 執行個體。應用程式層呼叫資料庫。

解決方案架構師應該如何提高傳輸中資料的安全性？

【選項】

- A. Configure a TLS listener. Deploy the server certificate on the NLB.
- B. Configure AWS Shield Advanced. Enable AWS WAF on the NLB.
- C. Change the load balancer to an Application Load Balancer (ALB). Enable AWS WAF on the ALB.
- D. Encrypt the Amazon Elastic Block Store (Amazon EBS) volume on the EC2 instances by using AWS Key Management Service (AWS KMS).

【參考答案】A

【題號】question 383

【題目】

A company is planning to migrate a commercial off-the-shelf application from its on-premises data center to AWS. The software has a software licensing model using

sockets and cores with predictable capacity and uptime requirements. The company wants to use its existing licenses, which were purchased earlier this year.

Which Amazon EC2 pricing option is the MOST cost-effective?

一家公司正計劃將商業現成應用程式從其本地資料中心遷移到 AWS。該軟體具有使用套接字和核心的軟體許可模型，具有可預測的容量和正常運行時間要求。該公司希望使用今年稍早購買的現有許可證。

哪種 Amazon EC2 定價選項最具成本效益？

【選項】

- A.Dedicated Reserved Hosts
- B.Dedicated On-Demand Hosts
- C.Dedicated Reserved Instances
- D.Dedicated On-Demand Instances

【參考答案】A

【題號】question 384

【題目】

A company runs an application on Amazon EC2 Linux instances across multiple Availability Zones. The application needs a storage layer that is highly available and Portable Operating System Interface (POSIX)-compliant. The storage layer must provide maximum data durability and must be shareable across the EC2 instances. The data in the storage layer will be accessed frequently for the first 30 days and will be accessed infrequently after that time.

Which solution will meet these requirements MOST cost-effectively?

一家公司在多個可用區的 Amazon EC2 Linux 執行個體上執行應用程式。應用程式需要一個高度可用且相容於可移植作業系統介面 (POSIX) 的儲存層。儲存層必須提供最大的資料持久性，並且必須可在 EC2 執行個體之間共用。儲存層的資料在前30天會被頻繁訪問，之後將不再被頻繁訪問。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A. Use the Amazon S3 Standard storage class. Create an S3 Lifecycle policy to move infrequently accessed data to S3 Glacier.

B. Use the Amazon S3 Standard storage class. Create an S3 Lifecycle policy to move infrequently accessed data to S3 Standard-Infrequent Access (S3 Standard-IA).

C. Use the Amazon Elastic File System (Amazon EFS) Standard storage class. Create a lifecycle management policy to move infrequently accessed data to EFS Standard-Infrequent Access (EFS Standard-IA).

D. Use the Amazon Elastic File System (Amazon EFS) One Zone storage class. Create a lifecycle management policy to move infrequently accessed data to EFS One Zone-Infrequent Access (EFS One Zone-IA).

【參考答案】C

【題號】question 385

【題目】

A solutions architect is creating a new VPC design. There are two public subnets for the load balancer, two private subnets for web servers, and two private subnets for MySQL. The web servers use only HTTPS. The solutions architect has already created a security group for the load balancer allowing port 443 from 0.0.0.0/0. Company policy requires that each resource has the least access required to still be able to perform its tasks.

Which additional configuration strategy should the solutions architect use to meet these requirements?

解決方案架構師正在創建新的 VPC 設計。有兩個用於負載平衡器的公有子網路、兩個用於 Web 伺服器的私有子網路和兩個用於 MySQL 的私有子網路。Web 伺服器僅使用 HTTPS。解決方案架構師已經為負載平衡器建立了一個安全性群組，允許來自 0.0.0.0/0 的連接埠 443。公司政策要求每個資源具有仍能夠執行其任務所需的最少存取權限。

解決方案架構師應該使用哪種附加配置策略來滿足這些要求？

【選項】

A.Create a security group for the web servers and allow port 443 from 0.0.0.0/0. Create a security group for the MySQL servers and allow port 3306 from the web servers security group.

B.Create a network ACL for the web servers and allow port 443 from 0.0.0.0/0. Create a network ACL for the MySQL servers and allow port 3306 from the web servers security group.

C.Create a security group for the web servers and allow port 443 from the load balancer. Create a security group for the MySQL servers and allow port 3306 from the web servers security group.

D.Create a network ACL for the web servers and allow port 443 from the load balancer. Create a network ACL for the MySQL servers and allow port 3306 from the web servers security group.

【參考答案】C

【題號】question 386

【題目】

An ecommerce company is running a multi-tier application on AWS. The front-end and backend tiers both run on Amazon EC2, and the database runs on Amazon RDS for MySQL. The backend tier communicates with the RDS instance. There are frequent calls to return identical datasets from the database that are causing performance slowdowns.

Which action should be taken to improve the performance of the backend?

一家電子商務公司正在 AWS 上運行多層應用程式。前端層和後端層均在 Amazon EC2 上運行，資料庫在 Amazon RDS for MySQL 上運行。後端層與 RDS 實例通訊。頻繁地呼叫從資料庫傳回相同的資料集會導致效能下降。

應該採取哪些措施來提高後端的效能？

【選項】

- A.Implement Amazon SNS to store the database calls.
- B.Implement Amazon ElastiCache to cache the large datasets.
- C.Implement an RDS for MySQL read replica to cache database calls.
- D.Implement Amazon Kinesis Data Firehose to stream the calls to the database.

【參考答案】B

【題號】question 387

【題目】

A new employee has joined a company as a deployment engineer. The deployment engineer will be using AWS CloudFormation templates to create multiple AWS resources. A solutions architect wants the deployment engineer to perform job activities while following the principle of least privilege.

Which combination of actions should the solutions architect take to accomplish this goal? (Choose two.)

一名新員工加入一家公司，擔任部署工程師。部署工程師將使用 AWS CloudFormation 範本建立多個 AWS 資源。解決方案架構師希望部署工程師在遵循最小特權原則的同時執行工作活動。

解決方案架構師應該採取哪些行動組合來實現這個目標？（選兩個。）

【選項】

- A.Have the deployment engineer use AWS account root user credentials for performing AWS CloudFormation stack operations.
- B.Create a new IAM user for the deployment engineer and add the IAM user to a group that has the PowerUsers IAM policy attached.
- C.Create a new IAM user for the deployment engineer and add the IAM user to a group that has the AdministratorAccess IAM policy attached.
- D.Create a new IAM user for the deployment engineer and add the IAM user to a group that has an IAM policy that allows AWS CloudFormation actions only.

E.Create an IAM role for the deployment engineer to explicitly define the permissions specific to the AWS CloudFormation stack and launch stacks using that IAM role.

【參考答案】DE

【題號】question 388

【題目】

A company is deploying a two-tier web application in a VPC. The web tier is using an Amazon EC2 Auto Scaling group with public subnets that span multiple Availability Zones. The database tier consists of an Amazon RDS for MySQL DB instance in separate private subnets. The web tier requires access to the database to retrieve product information.

The web application is not working as intended. The web application reports that it cannot connect to the database. The database is confirmed to be up and running. All configurations for the network ACLs, security groups, and route tables are still in their default states.

What should a solutions architect recommend to fix the application?

一家公司正在 VPC 中部署兩層 Web 應用程式。Web 層使用具有跨多個可用區的公有子網路的 Amazon EC2 Auto Scaling 群組。資料庫層由位於不同私有子網路中的 Amazon RDS for MySQL 資料庫執行個體所組成。Web 層需要存取資料庫來檢索產品資訊。

Web 應用程式未如預期運作。Web 應用程式報表無法連線到資料庫。資料庫已確認已啟動並正在運行。網路 ACL、安全性群組和路由表的所有設定仍處於預設狀態。

解決方案架構師應該建議什麼來修復應用程式？

【選項】

A.Add an explicit rule to the private subnet's network ACL to allow traffic from the web tier's EC2 instances.

B.Add a route in the VPC route table to allow traffic between the web tier's EC2 instances and the database tier.

C.Deploy the web tier's EC2 instances and the database tier's RDS instance into two separate VPCs, and configure VPC peering.

D.Add an inbound rule to the security group of the database tier's RDS instance to allow traffic from the web tiers security group.

【參考答案】D

【題號】question 389

【題目】

A company has a large dataset for its online advertising business stored in an Amazon RDS for MySQL DB instance in a single Availability Zone. The company wants business reporting queries to run without impacting the write operations to the production DB instance.

Which solution meets these requirements?

一家公司將其線上廣告業務的大型資料集儲存在單一可用區的 Amazon RDS for MySQL 資料庫執行個體中。該公司希望運行業務報告查詢而不影響對生產資料庫實例的寫入操作。

哪種解決方案滿足這些要求？

【選項】

A.Deploy RDS read replicas to process the business reporting queries.

B.Scale out the DB instance horizontally by placing it behind an Elastic Load Balancer.

C.Scale up the DB instance to a larger instance type to handle write operations and queries.

D.Deploy the DB instance in multiple Availability Zones to process the business reporting queries.

【參考答案】A

【題號】question 390

【題目】

A company hosts a three-tier ecommerce application on a fleet of Amazon EC2 instances. The instances run in an Auto Scaling group behind an Application Load Balancer (ALB). All ecommerce data is stored in an Amazon RDS for MariaDB Multi-AZ DB instance.

The company wants to optimize customer session management during transactions. The application must store session data durably.

Which solutions will meet these requirements? (Choose two.)

一家公司在一組 Amazon EC2 執行個體上託管一個三層電子商務應用程式。這些執行個體在應用程式負載平衡器 (ALB) 後面的 Auto Scaling 群組中執行。所有電子商務資料都儲存在 Amazon RDS for MariaDB 多可用區資料庫執行個體中。

該公司希望優化交易期間的客戶會話管理。應用程式必須持久儲存會話資料。

哪些解決方案可以滿足這些要求？（選兩個。）

【選項】

- A. Turn on the sticky sessions feature (session affinity) on the ALB.
- B. Use an Amazon DynamoDB table to store customer session information.
- C. Deploy an Amazon Cognito user pool to manage user session information.
- D. Deploy an Amazon ElastiCache for Redis cluster to store customer session information.
- E. Use AWS Systems Manager Application Manager in the application to manage user session information.

【參考答案】AD

【題號】question 391

【題目】

A company needs a backup strategy for its three-tier stateless web application. The web application runs on Amazon EC2 instances in an Auto Scaling group with a dynamic scaling policy that is configured to respond to scaling events. The database tier runs on Amazon RDS for PostgreSQL. The web application does not require temporary local storage on the EC2 instances. The company's recovery point objective (RPO) is 2 hours.

The backup strategy must maximize scalability and optimize resource utilization for this environment.

Which solution will meet these requirements?

一家公司需要為其三層無狀態 Web 應用程式制定備份策略。Web 應用程式在 Auto Scaling 群組中的 Amazon EC2 執行個體上執行，該群組具有配置為回應擴展事件的動態擴展策略。資料庫層在 Amazon RDS for PostgreSQL 上運作。Web 應用程式不需要 EC2 執行個體上的暫存本機。該公司的復原點目標 (RPO) 是 2 小時。

備份策略必須最大限度地提高可擴展性並優化該環境的資源利用率。

哪種解決方案可以滿足這些要求？

【選項】

A.Take snapshots of Amazon Elastic Block Store (Amazon EBS) volumes of the EC2 instances and database every 2 hours to meet the RPO.

B.Configure a snapshot lifecycle policy to take Amazon Elastic Block Store (Amazon EBS) snapshots. Enable automated backups in Amazon RDS to meet the RPO.

C.Retain the latest Amazon Machine Images (AMIs) of the web and application tiers. Enable automated backups in Amazon RDS and use point-in-time recovery to meet the RPO.

D.Take snapshots of Amazon Elastic Block Store (Amazon EBS) volumes of the EC2 instances every 2 hours. Enable automated backups in Amazon RDS and use point-in-time recovery to meet the RPO.

【參考答案】C

【題號】question 392

【題目】

A company wants to deploy a new public web application on AWS. The application includes a web server tier that uses Amazon EC2 instances. The application also includes a database tier that uses an Amazon RDS for MySQL DB instance.

The application must be secure and accessible for global customers that have dynamic IP addresses.

How should a solutions architect configure the security groups to meet these requirements?

一家公司想要在 AWS 上部署新的公共 Web 應用程式。該應用程式包括使用 Amazon EC2 執行個體的 Web 伺服器層。該應用程式還包括使用 Amazon RDS for MySQL 資料庫執行個體的資料庫層。

該應用程式必須安全且可供擁有動態 IP 位址的全球客戶存取。

解決方案架構師應如何配置安全群組以滿足這些要求？

【選項】

A. Configure the security group for the web servers to allow inbound traffic on port 443 from 0.0.0.0/0. Configure the security group for the DB instance to allow inbound traffic on port 3306 from the security group of the web servers.

B. Configure the security group for the web servers to allow inbound traffic on port 443 from the IP addresses of the customers. Configure the security group for the DB instance to allow inbound traffic on port 3306 from the security group of the web servers.

C. Configure the security group for the web servers to allow inbound traffic on port 443 from the IP addresses of the customers. Configure the security group for the DB instance to allow inbound traffic on port 3306 from the IP addresses of the customers.

D. Configure the security group for the web servers to allow inbound traffic on port 443 from 0.0.0.0/0. Configure the security group for the DB instance to allow inbound traffic on port 3306 from 0.0.0.0/0.

【參考答案】A

【題號】question 393

【題目】

A payment processing company records all voice communication with its customers and stores the audio files in an Amazon S3 bucket. The company needs to capture the text from the audio files. The company must remove from the text any personally identifiable information (PII) that belongs to customers.

What should a solutions architect do to meet these requirements?

一家支付處理公司記錄與其客戶的所有語音通信，並將音訊檔案儲存在 Amazon S3 儲存桶中。該公司需要從音訊檔案中捕獲文字。公司必須從文字中刪除屬於客戶的任何個人識別資訊 (PII)。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A.Process the audio files by using Amazon Kinesis Video Streams. Use an AWS Lambda function to scan for known PII patterns.

B.When an audio file is uploaded to the S3 bucket, invoke an AWS Lambda function to start an Amazon Textract task to analyze the call recordings.

C.Configure an Amazon Transcribe transcription job with PII redaction turned on. When an audio file is uploaded to the S3 bucket, invoke an AWS Lambda function to start the transcription job. Store the output in a separate S3 bucket.

D.Create an Amazon Connect contact flow that ingests the audio files with transcription turned on. Embed an AWS Lambda function to scan for known PII patterns. Use Amazon EventBridge to start the contact flow when an audio file is uploaded to the S3 bucket.

【參考答案】C

【題號】question 394

【題目】

A company is running a multi-tier ecommerce web application in the AWS Cloud. The application runs on Amazon EC2 instances with an Amazon RDS for MySQL Multi-AZ DB instance. Amazon RDS is configured with the latest generation DB instance with 2,000 GB of storage in a General Purpose SSD (gp3) Amazon Elastic Block Store (Amazon EBS) volume. The database performance affects the application during periods of high demand.

A database administrator analyzes the logs in Amazon CloudWatch Logs and discovers that the application performance always degrades when the number of read and write IOPS is higher than 20,000.

What should a solutions architect do to improve the application performance?

一家公司正在 AWS 雲端中運行多層電子商務 Web 應用程式。該應用程式在具有 Amazon RDS for MySQL 多可用區資料庫執行個體的 Amazon EC2 執行個體上執行。Amazon RDS 配置了最新一代資料庫實例，在通用 SSD (gp3) Amazon Elastic Block Store (Amazon EBS) 磁碟區中具有 2,000 GB 儲存空間。在高需求期間，資料庫效能會影響應用程式。

資料庫管理員分析 Amazon CloudWatch Logs 中的日誌，發現當讀寫 IOPS 高於 20,000 時，應用程式效能總是會下降。

解決方案架構師應該做什麼來提高應用程式效能？

【選項】

- A. Replace the volume with a magnetic volume.
- B. Increase the number of IOPS on the gp3 volume.
- C. Replace the volume with a Provisioned IOPS SSD (io2) volume.
- D. Replace the 2,000 GB gp3 volume with two 1,000 GB gp3 volumes.

【參考答案】D

【題號】question 395

【題目】

An IAM user made several configuration changes to AWS resources in their company's account during a production deployment last week. A solutions architect learned that a

couple of security group rules are not configured as desired. The solutions architect wants to confirm which IAM user was responsible for making changes.

Which service should the solutions architect use to find the desired information?

上週, 一位 IAM 用戶在生產部署期間對其公司帳戶中的 AWS 資源進行了多項配置變更。解決方案架構師了解到, 有些安全群組規則未如預期設定。解決方案架構師想要確認哪個 IAM 使用者負責進行變更。

解決方案架構師應該使用哪種服務來尋找所需的資訊?

【選項】

- A.Amazon GuardDuty
- B.Amazon Inspector
- C.AWS CloudTrail
- D.AWS Config

【參考答案】C

【題號】question 396

【題目】

A company has implemented a self-managed DNS service on AWS. The solution consists of the following:

- Amazon EC2 instances in different AWS Regions
- Endpoints of a standard accelerator in AWS Global Accelerator

The company wants to protect the solution against DDoS attacks.

What should a solutions architect do to meet this requirement?

一家公司在 AWS 上實施了自我管理的 DNS 服務。該解決方案包括以下內容:

- 不同AWS 區域中的Amazon EC2 執行個體
- AWS Global Accelerator 中標準加速器的終端節點

該公司希望保護該解決方案免受 DDoS 攻擊。

解決方案架構師應該怎麼做才能滿足這項要求？

【選項】

A.Subscribe to AWS Shield Advanced. Add the accelerator as a resource to protect.

B.Subscribe to AWS Shield Advanced. Add the EC2 instances as resources to protect.

C.Create an AWS WAF web ACL that includes a rate-based rule. Associate the web ACL with the accelerator.

D.Create an AWS WAF web ACL that includes a rate-based rule. Associate the web ACL with the EC2 instances.

【參考答案】A

【題號】question 397

【題目】

An ecommerce company needs to run a scheduled daily job to aggregate and filter sales records for analytics. The company stores the sales records in an Amazon S3 bucket. Each object can be up to 10 GB in size. Based on the number of sales events, the job can take up to an hour to complete. The CPU and memory usage of the job are constant and are known in advance.

A solutions architect needs to minimize the amount of operational effort that is needed for the job to run.

Which solution meets these requirements?

電子商務公司需要運行預定的日常作業來匯總和過濾銷售記錄以進行分析。該公司將銷售記錄儲存在 Amazon S3 儲存桶中。每個物件的大小最大可達 10 GB。根據銷售事件的數量，該作業可能需要長達一個小時才能完成。作業的 CPU 和記憶體使用量是恆定的且是預先知道的。

解決方案架構師需要最大限度地減少作業運行所需的操作工作量。

哪種解決方案滿足這些要求？

【選項】

A.Create an AWS Lambda function that has an Amazon EventBridge notification. Schedule the EventBridge event to run once a day.

B.Create an AWS Lambda function. Create an Amazon API Gateway HTTP API, and integrate the API with the function. Create an Amazon EventBridge scheduled event that calls the API and invokes the function.

C.Create an Amazon Elastic Container Service (Amazon ECS) cluster with an AWS Fargate launch type. Create an Amazon EventBridge scheduled event that launches an ECS task on the cluster to run the job.

D.Create an Amazon Elastic Container Service (Amazon ECS) cluster with an Amazon EC2 launch type and an Auto Scaling group with at least one EC2 instance. Create an Amazon EventBridge scheduled event that launches an ECS task on the cluster to run the job.

【參考答案】C

【題號】question 398

【題目】

A company needs to transfer 600 TB of data from its on-premises network-attached storage (NAS) system to the AWS Cloud. The data transfer must be complete within 2 weeks. The data is sensitive and must be encrypted in transit. The company's internet connection can support an upload speed of 100 Mbps.

Which solution meets these requirements MOST cost-effectively?

一家公司需要將 600 TB 的資料從其本地網路附加儲存 (NAS) 系統傳輸到 AWS 雲端。資料傳輸必須在 2 週內完成。資料很敏感，必須在傳輸過程中加密。該公司的網路連線可支援 100 Mbps 的上傳速度。

哪種解決方案最經濟有效率地滿足這些要求？

【選項】

A.Use Amazon S3 multi-part upload functionality to transfer the files over HTTPS.

B.Create a VPN connection between the on-premises NAS system and the nearest AWS Region. Transfer the data over the VPN connection.

C.Use the AWS Snow Family console to order several AWS Snowball Edge Storage Optimized devices. Use the devices to transfer the data to Amazon S3.

D.Set up a 10 Gbps AWS Direct Connect connection between the company location and the nearest AWS Region. Transfer the data over a VPN connection into the Region to store the data in Amazon S3.

【參考答案】C

【題號】question 399

【題目】

A financial company hosts a web application on AWS. The application uses an Amazon API Gateway Regional API endpoint to give users the ability to retrieve current stock prices. The company's security team has noticed an increase in the number of API requests. The security team is concerned that HTTP flood attacks might take the application offline.

A solutions architect must design a solution to protect the application from this type of attack.

Which solution meets these requirements with the LEAST operational overhead?

一家金融公司在 AWS 上託管 Web 應用程式。該應用程式使用 Amazon API Gateway Regional API 終端節點使用戶能夠檢索當前股票價格。該公司的安全團隊注意到 API 請求數量增加。安全團隊擔心 HTTP 洪水攻擊可能會使應用程式離線。

解決方案架構師必須設計一個解決方案來保護應用程式免受此類攻擊。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A.Create an Amazon CloudFront distribution in front of the API Gateway Regional API endpoint with a maximum TTL of 24 hours.

B.Create a Regional AWS WAF web ACL with a rate-based rule. Associate the web ACL with the API Gateway stage.

C. Use Amazon CloudWatch metrics to monitor the Count metric and alert the security team when the predefined rate is reached.

D. Create an Amazon CloudFront distribution with Lambda@Edge in front of the API Gateway Regional API endpoint. Create an AWS Lambda function to block requests from IP addresses that exceed the predefined rate.

【參考答案】B

【題號】question 400

【題目】

A meteorological startup company has a custom web application to sell weather data to its users online. The company uses Amazon DynamoDB to store its data and wants to build a new service that sends an alert to the managers of four internal teams every time a new weather event is recorded. The company does not want this new service to affect the performance of the current application.

What should a solutions architect do to meet these requirements with the LEAST amount of operational overhead?

一家氣象新創公司有一個客製化的 Web 應用程式，可以在線上向用戶出售天氣數據。該公司使用 Amazon DynamoDB 來儲存數據，並希望建立一項新服務，以便在每次記錄新的天氣事件時向四個內部團隊的經理發送警報。該公司不希望這項新服務影響目前應用程式的效能。

解決方案架構師應該如何做才能以最少的營運開銷來滿足這些要求？

【選項】

A. Use DynamoDB transactions to write new event data to the table. Configure the transactions to notify internal teams.

B. Have the current application publish a message to four Amazon Simple Notification Service (Amazon SNS) topics. Have each team subscribe to one topic.

C. Enable Amazon DynamoDB Streams on the table. Use triggers to write to a single Amazon Simple Notification Service (Amazon SNS) topic to which the teams can subscribe.

D.Add a custom attribute to each record to flag new items. Write a cron job that scans the table every minute for items that are new and notifies an Amazon Simple Queue Service (Amazon SQS) queue to which the teams can subscribe.

【參考答案】C

領先制勝